# PA2-2026 Report — FinSync (Group 04)

**Course:** CS300 - CSC13002 - Introduction to Software Engineering
**Assignment:** Project Assignment 2 (PA2-2026) — Sprint 2
**Group:** Group 04
**Project:** FinSync — Personal Finance & Group Expense Manager with AI Advisor
**Sprint 2 period:** `2026-10-10 to 2026-10-25`

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## Team Members

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

| # | Student ID | Full Name | Email | Main Role |
|---|---|---|---|---|
| 1 | 24120087 | Phạm Định Tiểu Long | phamlongkh2006@gmail.com | Project Manager / Group Leader / Business Analyst (BA) |
| 2 | 24120038 | Nguyễn Phú Đạt | nguyennphuudatt@gmail.com | UI/UX Designer & Frontend Lead |
| 3 | 24120403 | Nguyễn Lê Đức Nhật | nldnhat182006@gmail.com | Backend Lead |
| 4 | 24120342 | Vương Đắc Gia Khiêm | vuongkhiemvl10@gmail.com | QA Lead & DevOps |
| 5 | 24120051 | Ngô Thái Hòa | ngothaihoa235@gmail.com | AI Feature Lead & Documentation |

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## Part A — Project Plan

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

## 1. Introduction

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

FinSync is a mobile application that combines personal finance management (similar to *Money Lover*) with smart group expense sharing (similar to *Splitwise*) in a single product. It is designed for students, young professionals, and small groups who share costs such as rent, trips, and events.

The system works as a **bookkeeping, debt calculation, and reminder tool only**. It does not connect to real bank accounts and never moves money automatically. An AI Financial Advisor analyzes spending history and suggests a reasonable spending plan for the next month.

This document is the initial version of the Project Plan. It will be updated in PA3-2026 based on the TA feedback.

## 2. Project Overview

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

### 2.1 Goals

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

- Build a working mobile app that unifies personal and group finance tracking.
- Automate expense splitting and debt optimization to reduce the number of transfers between group members.
- Provide an AI Financial Advisor that turns raw spending data into clear monthly guidance.
- Provide a simple web admin panel for basic system administration.
- Apply Scrum and good software engineering practice throughout the semester (Jira, Git, code review, testing).

### 2.2 Scope

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

**In scope**

| # | Functional Group | Short Description |
|---|---|---|
| 1 | Authentication & Security | Register, login, logout, password recovery with JWT |
| 2 | Profile Management | Edit profile/password/avatar, Vietnamese/English interface, reporting currency (VND/USD) |

| # | Functional Group | Short Description |
|---|---|---|
| 3 | Wallet & Personal Account Management | Manual cash/bank/card wallets, original balances, converted net-worth overview |
| 4 | Personal Transaction Tracking | Online/offline income and expense entry, receipt capture, queued sync and conflict resolution |
| 5 | Group Management & Shared Wallets | Create groups, invite by link/code, Owner/Member roles, shared wallet |
| 6 | Smart Debt Split & Settlement | Exact splits, debt simplification, refunds, and external payments confirmed by the recipient |
| 7 | Budget Planning & Alerting | Weekly/monthly category budgets, alerts at 80% and 100% |
| 8 | Financial Reporting & Analytics | Pie/bar charts, group trip summary, PDF/Excel export |
| 9 | Savings Goal Management | Long-term goals and progress tracking |
| 10 | System Administration (Admin Panel) | User lock/unlock, default categories, reference exchange rates, and aggregate statistics |
| AI | AI Financial Advisor | Monthly analysis, anomaly detection, budget suggestions |

**Out of scope**

- Direct connection to real bank accounts or payment gateways.
- Automatic money deduction or real money transfer.
- iOS client.
- Executing currency exchange, automatic bank synchronization, and automatic financial transfers.
- Offline changes to group ledgers, settlement confirmations, or administrator data; offline group access is read-only.

### 2.3 Deliverables

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

| Deliverable | Sprint / PA | Format |
|---|---|---|
| Project Proposal, app survey | Sprint 1 / PA1 | Markdown + PDF |
| Project Plan (initial), Vision Document (initial), Spec Kit init | Sprint 2 / PA2 | Markdown + PDF, repo artifacts |
| Revised Project Plan and Vision Document after TA feedback | Sprint 3 / PA3 | Markdown + PDF |
| Use cases, architecture and design documents | Sprint 3–4 | Markdown + PDF |

| Deliverable | Sprint / PA | Format |
|---|---|---|
| Test plan and test cases | Sprint 4 | Markdown + PDF |
| Source code: mobile app, backend API, AI microservice, admin web | Sprint 2–5 | GitHub repository (private) |
| Test builds | See Build Plan | APK / deployed environment |
| Final demo of all features | Sprint 5 / PA5 | Live demo |

### 2.4 Assumptions

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

- Every member works as a full-stack engineer; roles below only define the main person in charge (Lead).
- Each of the five members allocates at least **10 hours per week**, equivalent to approximately **114 team hours** over the 16-calendar-day PA2 window. Work is distributed between individual outputs, review, and meetings.
- Local Docker environments are the baseline for development and rehearsal. Remote hosting is optional for the demo; external AI calls are limited, cached where appropriate, and controlled by configuration so development can continue with synthetic responses during provider outages.
- Users enter all transactions manually; actual payments happen outside the app.
- Dates for PA3–PA5 are internal planning targets in this initial plan; official course deadlines take precedence. Their detailed scope will be refined at the start of each sprint.

## 3. Project Organization

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Nguyen Le Duc Nhat` | *Edited by:* `Pham Dinh Tieu Long`

### 3.1 Team Structure and Roles

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Nguyen Le Duc Nhat` | *Edited by:* `Pham Dinh Tieu Long`

| Member | Role | Main Responsibilities |
|---|---|---|
| Phạm Định Tiểu Long | Project Manager / Group Leader / Business Analyst (BA) | Track progress, coordinate work, lead Sprint meetings, and manage the Jira board; elicit and clarify user needs, analyze business rules, define requirements and acceptance criteria with the team, and maintain consistency between the Vision Document and backlog |

| Member | Role | Main Responsibilities |
|---|---|---|
| Nguyễn Phú Đạt | UI/UX Designer & Frontend Lead | Wireframes and mockups, UI consistency, lead mobile client development |
| Nguyễn Lê Đức Nhật | Backend Lead | Backend architecture, Spring Boot REST API, database design |
| Vương Đắc Gia Khiêm | QA Lead & DevOps | Testing strategy, test cases, CI/CD pipeline, deployment |
| Ngô Thái Hòa | AI Feature Lead & Documentation | AI Advisor (Gemini/OpenAI integration), technical documentation |

Following TA feedback after PA1 that the team lacked a Business Analyst role, Phạm Định Tiểu Long also takes responsibility for business analysis from PA2 onward, alongside his Project Manager and Group Leader duties. He coordinates requirements clarification with the team and supports feature authors and reviewers in resolving scope and business-rule questions.

**Working rules**

- Each Jira task is assigned to exactly **one** member and reviewed by **another** member.
- Tasks are created **before** work starts (never created, assigned, and closed at the same time).
- Commit messages follow Conventional Commits: `<type>(<scope>): <subject>`.
- Secrets (API keys, passwords, JWT secret, `.env`) are never committed. The repository is private.

### 3.2 Risk Management

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Nguyen Le Duc Nhat` | *Edited by:* `Pham Dinh Tieu Long`

| ID | Risk | Probability | Impact | Mitigation Strategy | Owner |
|---|---|---|---|---|---|
| R1 | Member unavailability or uneven workload | Medium | High | Limit most work packages to three days, reserve review time, use a different reviewer as backup, and redistribute incomplete work during Scrum. | Long |
| R2 | Scope creep across eleven feature groups | High | High | Deliver a basic usable flow for every planned feature by Build 3; prioritize correctness and integration, and defer cosmetic depth through a documented scope decision. | Long |
| R3 | Learning Kotlin/Compose, Spec Kit, and deployment | Medium | Medium | Schedule individual training and small runnable | Đạt |

| ID | Risk | Probability | Impact | Mitigation Strategy | Owner |
|---|---|---|---|---|---|
| | tools | | | skeletons in Sprint 2; retain the PA1 stack and share setup instructions. | |
| R4 | AI quota, latency, cost, or invalid suggestions | Medium | Medium | Use bounded requests, aggregate input, schema validation, a 30-second timeout, cached advice, and synthetic development responses. Keep finance functions independent of the AI provider. | Hòa |
| R5 | Mobile/backend/AI/admin integration failures | Medium | High | Define API contracts and shared currency/accounting rules early, smoke-test each build, and integrate before the final review. | Nhật |
| R6 | Exposed credentials or unauthorized financial access | Low | High | Externalize secrets, review tracked files and authorization tests, limit AI payloads, and rotate any exposed credential. | Khiêm |
| R7 | Incorrect shares, refunds, or settlements | Medium | High | Use exact money arithmetic, deterministic remainder allocation, immutable adjustments, atomic updates, and recipient-confirmation/retry tests. | Nhật |
| R8 | Lost or duplicated offline changes | Medium | High | Persist a local queue, identify operations uniquely, retain conflicting versions, test restart/retry behavior, and keep shared writes online. | Đạt |
| R9 | Stale or inconsistent exchange rates | Medium | Medium | Store rate/source/time with converted entries, label cached rates, reject missing-rate conversions, and test that rate updates never rewrite historical debt. | Nhật |

## 4. Project Plan

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Vuong Dac Gia Khiem` | *Edited by:* `Pham Dinh Tieu Long`

### 4.1 Process

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Vuong Dac Gia Khiem` | *Edited by:* `Pham Dinh Tieu Long`

The project follows the **Scrum** process. The semester is divided into **5 sprints**, each matching one Project Assignment (Sprint 1 = PA1, Sprint 2 = PA2, and so on). Each sprint lasts 2–3 weeks and has four meetings:

| Meeting | When | Purpose |
|---|---|---|
| Sprint Planning (1) | At the transition before sprint implementation; Sprint 2 kickoff slot: 10 October, 09:00–10:00 | Prioritize work, allocate capacity, and assign tasks |
| Weekly Scrum (2) | During sprint | Each member answers: done since last week / plan until next week / obstacles |
| Sprint Review & Retrospective (1) | End of sprint | Review results; what went well, what went wrong, root causes, improvements, lessons learned |

Tools follow the PA1 team contract: Jira (task tracking), GitHub (code and review), Discord / Zalo (daily communication), and Google Meet / Zoom (meetings).

The Jira workflow is **To Do** → **In Progress** → **Done**, as recorded in PA1. Work remains In Progress until another member reviews it. A task is Done when its output is linked, its acceptance criteria are checked, its review is recorded, and its actual completion date is entered. Documentation and self-training tasks follow the same rules as coding tasks.

### 4.2 Sprint Overview

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Vuong Dac Gia Khiem` | *Edited by:* `Pham Dinh Tieu Long`

| Sprint | PA | Planning period | Main focus |
|---|---|---|---|
| Sprint 1 | PA1 | Before 2026-10-10; PA1 artifacts available in repository by 2026-10-05 | Proposal, app survey, team contract, and initial repository/backend setup |
| Sprint 2 | PA2 | **2026-10-10 to 2026-10-25** | Initial plan/vision, Spec Kit/training, skeletons, and Build 0 |
| Sprint 3 | PA3 | 2026-10-26 to 2026-11-08 | Revised documents; Auth, Profile, Wallet, offline Personal Transactions; AI/admin prototypes; |

| Sprint | PA | Planning period | Main focus |
|---|---|---|---|
| | | | Build 1 |
| Sprint 4 | PA4 | 2026-11-09 to 2026-11-22 | Groups, shared fund, split/settlement/refunds, currency conversion, budgets, savings, reports; Build 2 |
| Sprint 5 | PA5 | 2026-11-23 to 2026-12-06 | Complete AI/admin/export, full regression and offline rehearsal, release candidate, final build, and demo |

Sprint 3–5 dates are team planning targets aligned to two-week cycles, rather than confirmed course submission deadlines. Sprint Planning refines each later sprint's task-level estimates and assignments.

### 4.3 Sprint 2 — Detailed Task List

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Vuong Dac Gia Khiem` | *Edited by:* `Pham Dinh Tieu Long`

The dated task list is the Sprint 2 baseline. Every task has one assignee and a different reviewer; due dates include the owner/reviewer handoff. The S2 identifiers name work packages and are retained in Jira task titles. Section D.2.5 contains the single board screenshot and the four Jira-key mappings readable in that image. The baseline dates below remain unchanged for comparison with the recovery targets recorded in D.2.2: individual training by 20 October and completion of the Spec Kit constitution/inventory by 21 October. Long owns S2-12 and Nhật reviews it. Part C records initial setup work on 19 October; this date does not imply completion of the entire constitution and inventory task. The meetings took place on 10, 14, 18, and 22 October. All assigned work was completed by the 22 October Review, ahead of the original 25 October endpoint. Non-meeting task dates below show the original baseline.

Long, Đạt, Nhật, Khiêm, and Hòa refer to the members in the team table. Meetings involve all five members; the assigned owner facilitates and records the minutes. Estimates below cover the owner's task execution; peer review and other members' meeting attendance are budgeted separately.

| Local ID | Task | Assignee | Reviewer | Planned start | Due date | Owner effort (h) |
|---|---|---|---|---|---|---|
| S2-01 | Sprint 2 Planning and meeting notes | Long | Hòa | 2026-10-10 | 2026-10-10 | 1.5 |
| S2-02 | Project Plan — Introduction and Overview | Nhật | Long | 2026-10-13 | 2026-10-14 | 2 |
| S2-03 | Project Plan — Organization and Risk Management | Long | Khiêm | 2026-10-13 | 2026-10-14 | 2 |
| S2-04 | Project Plan — Schedule and Build Plan | Long | Khiêm | 2026-10-15 | 2026-10-16 | 2 |

| Local ID | Task | Assignee | Reviewer | Planned start | Due date | Owner effort (h) |
|---|---|---|---|---|---|---|
| S2-05 | Vision — Introduction and Positioning | Hòa | Long | 2026-10-13 | 2026-10-14 | 2 |
| S2-06 | Vision — Users, Stakeholders, Alternatives; review PA1 survey | Đạt | Hòa | 2026-10-13 | 2026-10-15 | 3 |
| S2-07 | Vision — Product Overview and accounting boundaries | Nhật | Khiêm | 2026-10-15 | 2026-10-16 | 3 |
| S2-08 | Vision — Product Features, 3–5 sentences per feature | Hòa | Nhật | 2026-10-15 | 2026-10-17 | 4 |
| S2-09 | Two Mermaid user workflows | Đạt | Khiêm | 2026-10-18 | 2026-10-19 | 3 |
| S2-10 | Vision — NFR targets and verification methods | Khiêm | Nhật | 2026-10-18 | 2026-10-19 | 3 |
| S2-11 | Review existing GitHub setup, folders, ignore rules, and branch rules | Khiêm | Long | 2026-10-10 | 2026-10-11 | 1.5 |
| S2-12 | Initialize Spec Kit in `/src`; complete constitution and artifact inventory | Long | Nhật | 2026-10-13 | 2026-10-15 | 3 |
| S2-13 | Spec Kit training and individual evidence — Long | Long | Hòa | 2026-10-10 | 2026-10-12 | 3 |
| S2-14 | Spec Kit training and individual evidence — Đạt | Đạt | Long | 2026-10-10 | 2026-10-12 | 3 |
| S2-15 | Spec Kit training and individual evidence — Nhật | Nhật | Đạt | 2026-10-10 | 2026-10-12 | 3 |
| S2-16 | Spec Kit training and individual evidence — Khiêm | Khiêm | Nhật | 2026-10-10 | 2026-10-12 | 3 |
| S2-17 | Spec Kit training and individual evidence — Hòa | Hòa | Khiêm | 2026-10-10 | 2026-10-12 | 3 |
| S2-18 | Kotlin / Compose self-training and runnable mobile skeleton | Đạt | Nhật | 2026-10-16 | 2026-10-18 | 4 |
| S2-19 | Review/complete backend skeleton, database, migrations, and API documentation | Nhật | Khiêm | 2026-10-17 | 2026-10-19 | 6 |

| Local ID | Task | Assignee | Reviewer | Planned start | Due date | Owner effort (h) |
|---|---|---|---|---|---|---|
| S2-20 | AI service self-training, skeleton, connectivity and failure smoke test | Hòa | Nhật | 2026-10-18 | 2026-10-20 | 4 |
| S2-21 | Weekly Scrum #1 and notes | Long | Hòa | 2026-10-14 | 2026-10-14 | 1 |
| S2-22 | Weekly Scrum #2 and notes | Long | Hòa | 2026-10-18 | 2026-10-18 | 1 |
| S2-23 | Sprint Review & Retrospective and notes | Long | Khiêm | 2026-10-22 | 2026-10-22 | 1.5 |
| S2-24 | Consolidate AI usage records and prompt histories | Hòa | Long | 2026-10-21 | 2026-10-23 | 2 |
| S2-25 | Jira/Git evidence, Markdown/PDF visual check, final submission package | Khiêm | Long | 2026-10-24 | 2026-10-25 | 3 |
| S2-26 | Produce Build 0 and record environment smoke-test results | Khiêm | Đạt | 2026-10-21 | 2026-10-22 | 2 |
| S2-27 | Cross-check report against PA2, resolve reviews, and reconcile Jira | Đạt | Hòa | 2026-10-23 | 2026-10-24 | 2 |

**Capacity allocation**

| Member | Owned tasks (h) | Peer review (h) | Additional meeting attendance (h) | Planned total (h) |
|---|---|---|---|---|
| Long | 15 | 3 | 0 | 18 |
| Đạt | 15 | 2.5 | 3.5 | 21 |
| Nhật | 14 | 3 | 3.5 | 20.5 |
| Khiêm | 12.5 | 2.5 | 3.5 | 18.5 |
| Hòa | 15 | 2.5 | 3.5 | 21 |

The plan allocates **99 team hours** from approximately **114 available hours**, leaving about **15 hours** for corrections, environment problems, and coordination. Long's meeting tasks already include his attendance and minutes preparation; the other four members each reserve 3.5 hours for attendance.

**Dependencies and checkpoints:** repository checks finish by 11 October and individual training by 12 October, before Spec Kit setup on 13–15 October. Feature/accounting descriptions precede the 18–19

October workflows and NFRs. Mobile, backend, and AI skeletons are due by 20 October, before Build 0 on 21–22 October. The original baseline reserved 23–24 October for corrections and 25 October for packaging. Actual completion was reported at the 22 October Review.

**Definition of Done:** the output exists, meets its acceptance criteria, has a different member's recorded review, and has accurate actual dates/status in Jira. A training task includes individual learning evidence; a meeting task includes attendance, updates, decisions, and actions; a build task includes reproducible startup steps and smoke-test results. The single Jira board screenshot in Section D.2.5 summarizes task assignments and progress. Completion is confirmed through the task output and recorded review.

### 4.4 Build Plan

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Vuong Dac Gia Khiem` | *Edited by:* `Pham Dinh Tieu Long`

The final sprint (PA5) includes a demo of all features, so builds are planned to deliver features gradually.

| Build | Target date | Planned content | Acceptance focus |
|---|---|---|---|
| Build 0 — skeleton | 2026-10-22 | Runnable mobile/backend/database/AI skeletons and startup instructions | Environment reproduction, service connectivity, failure handling, and recorded smoke tests |
| Build 1 | 2026-11-08 | Auth, Profile, wallets, offline personal transactions and synchronization; AI/admin prototypes | Access control, restart/retry safety, conflict handling, and core transaction flows |
| Build 2 | 2026-11-22 | Groups/shared fund, splits/refunds, confirmed settlements, VND/USD conversion, budgets, savings, reports | Exact financial totals, historical rates, group permissions, and connected/offline boundaries |
| Build 3 — release candidate | 2026-11-29 | All ten functional groups and AI Advisor, complete admin/rate tools, PDF/Excel export | End-to-end feature coverage, security/performance tests, and 24-hour offline rehearsal |
| Final build | 2026-12-04 | Reviewed fixes to the release candidate | Regression closure, complete runbook, final data/evidence, and demo rehearsal |

**Five testable builds** are planned. Each build is associated with a source revision/tag, installation/startup steps, test outcomes, and known issues. Build 3 delivers all planned feature groups before final regression; the final sprint's internal demo target is **6 December 2026**, with 5–6 December reserved for rehearsal and presentation. Dates are adjusted if official PA3–PA5 deadlines require it.

### 4.5 Plan for Later Sprints (planned tasks only)

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Vuong Dac Gia Khiem` | *Edited by:* `Pham Dinh Tieu Long`

**Sprint 3 (PA3)**

- Revise Project Plan and Vision Document based on TA feedback.
- Write use cases and requirements specification (`/speckit.specify`).
- Design database schema, API contracts, and UI mockups.
- Implement Authentication, Profile, Wallet, and Personal Transactions, including encrypted offline storage, queued changes, synchronization, and conflict handling.
- Prototype the AI analysis contract with synthetic data and the admin login/user-list flow.
- Write test plan; produce Build 1.

**Sprint 4 (PA4)**

- Implement online Group Management, confirmed shared-wallet contributions, exact splitting/refunds, recipient-confirmed settlement, and auditable adjustments.
- Implement VND/USD conversion with stored rate snapshots, Budget Planning & Alerting, Savings Goals, Reports, and charts.
- Integrate the first AI analysis and admin management flows; exercise timeout and permission handling.
- Write test cases; set up CI pipeline; produce Build 2.

**Sprint 5 (PA5)**

- Implement AI Financial Advisor and Admin Panel; export PDF/Excel.
- Run integration, financial-invariant, performance/security, and 24-hour offline tests; fix defects and verify backup/restore.
- Prepare final report and demo; produce Build 3 and the final build.

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# Part B — Vision Document

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

## 1. Introduction

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

### 1.1 Purpose

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

This document describes the vision for **FinSync**: the problem it solves, the people who will use it, its main features, and the quality requirements it must meet. It provides a common reference for the team, lecturer, and TAs when prioritizing work and evaluating the final product. The PA2 vision includes offline personal transaction management, VND/USD conversion, and explicit rules for shared expenses and settlement. This initial version will be revised in PA3-2026 based on TA feedback.

### 1.2 References

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

| Reference | Document and use |
|---|---|
| R1 | PA1 Project Proposal — Part B: original product scope, target users, and feature baseline |
| R2 | PA1 Existing App Survey — Part C: unchanged Money Lover and Splitwise findings and screenshots |
| R3 | PA1 Team Contract — Part D: team roles and working agreements |
| R4 | PA2 Assignment Specification: required Project Plan, Vision Document, and submission format |
| R5 | FinSync Project Context: system boundaries, technologies, and business domains |
| R6 | Part A of this report: sprint schedule, responsibilities, dependencies, and build plan |

## 2. Positioning

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

### 2.1 Problem Statement

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

| | |
|---|---|
| **The problem of** | In the team's PA1 survey, personal tracking and group splitting were handled by separate tools. Users must combine the information by hand, and often forget small debts. |
| **Affects** | Students, young professionals, and small groups (roommates, travel friends, project teams) who spend money together and also want to control their own spending. |
| **The impact of which is** | Time wasted on manual calculation, unclear "who owes whom", conflicts between friends, an incomplete view of personal spending, and overspending without early warning. |
| **A successful solution would be** | One app that combines personal wallets and budgets with group expense splitting and debt optimization, and gives clear AI-based advice on how to spend in the next month. |

### 2.2 Product Position Statement

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

| | |
|---|---|
| **For** | Students and young professionals who manage personal money and share costs with others |
| **Who** | Need to track personal spending, split shared bills fairly, and plan their monthly budget |
| **The product name** | FinSync |
| **That** | Combines personal finance tracking, smart group expense splitting with debt optimization, and an AI Financial Advisor in one mobile app |
| **Unlike** | Using separate personal-finance and group-splitting tools, represented by Money Lover and Splitwise in the PA1 survey |
| **Our product** | Unifies personal and group finance in one place, reduces the number of transfers needed to settle debts, and turns spending history into clear monthly advice — without touching the user's real bank accounts |

## 3. Stakeholder and User Descriptions

> *Performed by:* `Ngo Thai Hoa` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Ngo Thai Hoa`

### 3.1 Stakeholder Summary

> *Performed by:* `Ngo Thai Hoa` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Ngo Thai Hoa`

| Stakeholder | Description | Main Interest |
|---|---|---|
| Regular Users | Students and young professionals using the app | Easy tracking, fair splitting, useful advice |
| System Administrator | Person managing users and default data | Simple tools to manage accounts and categories |
| Development Team (Group 04) | Five students building the system | Deliver a working product on time with good quality |
| Lecturer and TAs | Course staff who grade and give feedback | Correct process, documents, and working software |
| Third-party services (Gemini/OpenAI, cloud hosting) | Providers of AI and hosting | Fair use within limits and terms |

### 3.2 User Summary

> *Performed by:* `Ngo Thai Hoa` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Ngo Thai Hoa`

| User | Description | Responsibilities in the System |
|---|---|---|
| **Regular User** | Individual who tracks money and joins expense groups | Manage wallets, transactions, budgets, goals; create or join groups; confirm payments |
| ↳ Group Owner | Regular User who creates and manages a group | Invite members, manage group settings and group wallet |
| ↳ Group Member | Regular User who joins a group | Add shared expenses, view debts, mark payments |
| **Administrator** | Manages the system through a simple web page | View users, lock/unlock accounts, manage default categories, view statistics |

### 3.3 User Environment

> *Performed by:* `Ngo Thai Hoa` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Ngo Thai Hoa`

- **Regular Users** use an Android smartphone, as selected in PA1, on mobile data or Wi-Fi, often in short sessions for recording an expense. iOS is outside the current baseline unless a scope change is approved.
- **Offline support is included.** After an initial online sign-in and synchronization, users can view cached wallets, transactions, budgets, goals, and reports, and create, edit, or delete personal transactions without a connection. The app stores these changes locally, shows their pending status, and synchronizes them when connectivity and a valid session are available.
- **Shared operations require a connection.** Group creation/invitations, shared expense changes, settlement confirmation, new AI analysis, and administration use the latest server state. Cached group balances and previous AI advice remain readable offline with their last-sync time; the app explains when an action requires a connection.
- Receipt images captured offline remain on the device and upload after reconnection. Updates to the same personal record on two devices are shown for resolution rather than silently overwriting either version.
- **Administrators** use a desktop web browser (Chrome, Edge, Firefox).
- **Currencies and language:** VND and USD are supported with bidirectional conversion; Vietnamese and English are available in the user interface. A user's selected reporting currency determines converted summaries, while every record retains its original currency and amount.

### 3.4 Summary of Key Stakeholder or User Needs

> *Performed by:* `Ngo Thai Hoa` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Ngo Thai Hoa`

| Need | Priority | Current Solution | Proposed Solution |
|---|---|---|---|
| Record personal income/expense quickly, including offline | High | Notes app, spreadsheets, Money Lover | Local transaction entry with categories, receipt capture, visible sync status, and synchronization after reconnection |

| Need | Priority | Current Solution | Proposed Solution |
|---|---|---|---|
| See total money across cash, bank, and card in one reporting currency | High | Manual calculation | Wallet dashboard with VND/USD conversion and visible exchange-rate details |
| Split shared bills fairly | High | Mental calculation, chat messages, Splitwise | Equal, percentage, and exact splits |
| Know who owes whom and settle with few transfers | High | Manual reminders | Debt summary, simplified repayment suggestions, reminders, and settlement confirmed by the recipient |
| Avoid overspending | Medium | Self-control | Category budgets with alerts at 80% and 100% |
| Save for long-term goals | Medium | Notes, memory | Savings goals with progress tracking |
| Understand spending and plan next month | Medium | Manual analysis | AI Financial Advisor with monthly suggestions |
| Keep financial data private and safe | High | Depends on the app | Authentication, JWT, private personal data |
| Manage users and default categories | Medium | None | Web admin panel |

### 3.5 Alternatives and Competition

> *Performed by:* `Ngo Thai Hoa` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Ngo Thai Hoa`

The team retains the Money Lover and Splitwise survey from PA1 Part C without changes to its observations, comparison results, or supporting screenshots. The table below reproduces the feature comparison in PA1 Part C.3. Competitor entries describe the PA1 survey baseline; FinSync entries describe the planned product.

| Feature | Money Lover | Splitwise | FinSync (planned) |
|---|---|---|---|
| Personal wallet management | ✅ | ❌ | ✅ |
| Personal transaction tracking | ✅ | ❌ | ✅ |
| Budget planning & alerts | ✅ | ❌ | ✅ |
| Financial reports & charts | ✅ | Limited | ✅ |
| Savings goals | ✅ | ❌ | ✅ |
| Group expense management | ❌ | ✅ | ✅ |
| Smart debt splitting | ❌ | ✅ | ✅ |

| Feature | Money Lover | Splitwise | FinSync (planned) |
|---|---|---|---|
| Debt simplification algorithm | ❌ | ✅ | ✅ |
| Payment reminders | ❌ | ✅ | ✅ |
| AI financial advisor | ❌ | ❌ | ✅ |
| Admin panel | ❌ | ❌ | ✅ |
| Unified personal + group finance | ❌ | ❌ | ✅ |

Money Lover provides the personal dashboard, category selection, wallet aggregation, and reporting patterns that inform FinSync's personal-finance experience. Splitwise provides the group balance, flexible split, and external-settlement patterns that inform shared-expense workflows. FinSync's planned differentiation remains the combination of personal and group finance with monthly AI guidance and a Vietnamese-friendly interface. Offline operation and currency conversion are specified as FinSync requirements here; no additional competitor findings are asserted.

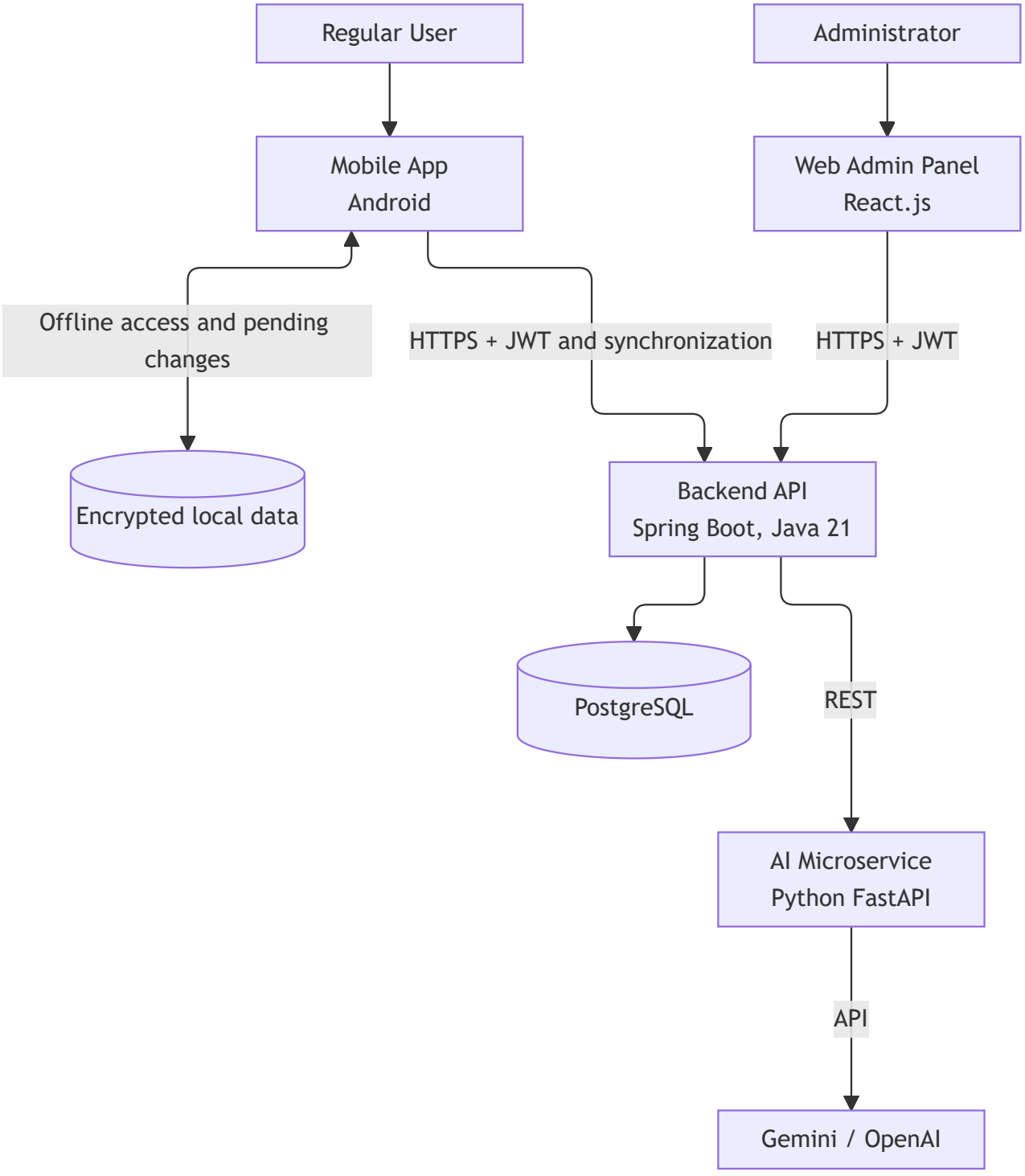
## 4. Product Overview

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Nguyen Le Duc Nhat` | *Edited by:* `Pham Dinh Tieu Long`

### 4.1 Product Perspective

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Nguyen Le Duc Nhat` | *Edited by:* `Pham Dinh Tieu Long`

FinSync is a new, self-contained product made of four parts that communicate through REST APIs:




The product is a **bookkeeping, calculation, and reminder** system. Payments happen outside the app (bank transfer, cash), and users confirm them in the app. The product never accesses real bank accounts.

The system separates two business areas:

- **Personal Expense Management:** private data, visible only to the account owner.
- **Group Expense Management:** shared data, visible to members of the same group.

**Accounting and reconciliation rules**

| Rule | Required behavior |
| --- | --- |
| Group permissions | The Owner manages membership and settings; members may add expenses and payment claims. An expense recorder or the Owner may correct an expense, with the original values retained in history. Changes affecting previously settled amounts use linked adjustments and recalculated balances. |
| Personal and group ownership | Only the owner can access personal records. Group records are shared only with current authorized members; administrative statistics do not expose individual private transaction histories. |
| Expense recognition | Personal spending reports, category budgets, and AI analysis include only the user's own share of a group expense. The full amount advanced by a member is a cash movement; the amount recoverable from others is tracked as a receivable rather than additional personal consumption. |
| Linked records | A personal wallet movement, group expense, and allocated share may reference the same event. Reports identify that event and count consumption once, including when a user links an existing personal entry to a group expense. |
| Shared-wallet contributions | A member records a contribution claim and the Owner, acting as fund custodian, confirms receipt before it increases the shared fund and contributor's fund balance; it is a transfer, not an expense. An expense paid from the fund reduces available cash and allocates consumption to participants. Differences between each member's contributions and allocated costs determine amounts owed to or recoverable from the fund. A direct payment by a member is recorded against that member instead, so both funding methods cannot fund the same expense. |
| Shared-wallet visibility | The group fund balance is shown separately from personal wallets. Its entire balance is never included as one member's personal asset; member contribution and reimbursement positions are shown separately. All balances represent bookkeeping entries. |
| Splitting and rounding | Equal, percentage, and exact-amount splits operate in the group's currency. Percentages must total 100%; exact shares must equal the expense. Any rounding remainder is assigned in a stable member order so shares always sum to the recorded total. |
| Settlement authorization | The debtor records a positive payment claim after paying outside the app; the intended recipient confirms receipt before the debt is reduced. A rejected or pending claim leaves the confirmed debt unchanged. Partial settlement is allowed up to the outstanding amount; confirming the same claim twice has no additional effect. An Owner cannot confirm on another recipient's behalf. |
| Settlement accounting | Confirmed settlement changes debt and any linked cash movement, not consumption or income. Linking a previously recorded payment prevents a second cash entry. A return of unused shared funds follows the same sender/recipient confirmation rule and is a transfer, not an expense. |

| Rule | Required behavior |
| --- | --- |
| Refunds and corrections | Refunds reference the original expense, cannot exceed its unrefunded amount, and reverse the original participants' shares proportionally with exact rounding. Previously confirmed payments remain in history; resulting overpayments create reimbursement balances. Changes affecting balances are recorded as linked adjustments rather than silently deleting settled history. |
| Server reconciliation | Financial updates are applied atomically, preserve an audit trail, and reject stale conflicting versions. Retrying an operation with the same identifier cannot duplicate a transaction, share, refund, or settlement. Online group debt and fund calculations use the same ledger and currency rules. |

**Currency conversion rules**

FinSync supports conversion in both directions between VND and USD. Users select VND or USD as their reporting currency; each wallet and group has its own currency, which cannot be changed after financial entries exist.

| Item | Conversion rule |
| --- | --- |
| Exchange-rate source | Administrators maintain a shared reference-rate table for the course deployment, recording a positive VND-per-USD rate, the quoted source, effective date/time, and update author. Rate entry is manual; a live exchange-rate API is not required for this version. |
| Conversion direction | USD amounts are multiplied by the recorded VND-per-USD rate to obtain VND; VND amounts are divided by that rate to obtain USD. The original amount is retained. |
| Precision | Monetary values use exact decimal arithmetic. VND is rounded to whole dong and USD to two decimal places using half-up rounding; conversion is performed before group shares are allocated. |
| Transaction history | Every converted financial entry retains the applied rate, source, and effective timestamp. Later rate updates do not change previously recorded expense shares or confirmed settlements. |
| Cross-currency group entries | An expense entered in another currency is converted once into the group's currency using the recorded rate snapshot. Debts and settlement limits are calculated in the group's currency; the original amount remains visible for checking. |
| Summaries and reports | Personal net-worth summaries show original wallet balances and converted equivalents using an identified valuation rate. Historical spending reports use transaction-rate snapshots and state their reporting currency. Changing the reporting preference recalculates the presentation without rewriting the ledger. |
| Offline and unavailable rates | Offline screens may use the last synchronized rate and show its effective date and offline status. If a rate is unavailable, original-currency records can still be saved, |

| Item | Conversion rule |
| --- | --- |
| | while conversion-dependent totals or shared operations wait for a valid rate. Reconnection does not silently replace a saved transaction's applied rate. |

**Offline synchronization**

The Android client keeps an encrypted local copy of the signed-in user's permitted data. Personal transaction changes receive stable identifiers and remain visible with pending, synchronized, or conflict status. Synchronization preserves transaction dates and prevents duplicate retries; version conflicts retain both versions for explicit user resolution. Group data remains a read-only cached snapshot while offline, and a valid online session is required before pending personal changes are submitted.

### 4.2 Assumptions and Dependencies

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Nguyen Le Duc Nhat` | *Edited by:* `Pham Dinh Tieu Long` **Assumptions**

- Users enter transactions manually and honestly.
- Users have an Android smartphone and can connect for initial sign-in, synchronization, and shared operations; personal transaction entry remains available during temporary disconnection.
- Group members agree on expense participants, contributions, and refunds, and confirm actual external payments in the app.
- Users understand that converted totals are bookkeeping estimates using displayed reference rates; the application does not execute currency exchange or bank transfers.

**Dependencies**

| Dependency | Used For | Risk |
| --- | --- | --- |
| Google Gemini / OpenAI API | AI Financial Advisor | Quota, cost, availability |
| PostgreSQL | Data storage | Hosting and backup |
| Docker / Docker Compose | Local and deployment environment | Setup problems |
| Docker-capable Linux host | Backend, PostgreSQL, AI service, and admin deployment | Resource limits and host availability; a local Docker environment supports rehearsal |
| In-app notification store | Debt reminders, settlement requests, and budget alerts | Offline users see server-side notifications after the next successful sync; local budget warnings remain available |
| Administrator-maintained exchange-rate table | VND/USD conversion with recorded rate/source/time | Stale or missing rates are shown explicitly and never silently invented |

| Dependency | Used For | Risk |
|---|---|---|
| Android encrypted local storage and sync queue | Offline personal transactions and cached authorized data | Device storage, session renewal, and explicit resolution of conflicting edits |
| Transactional email service | Password recovery using expiring, single-use reset links | Delivery failure or delay; recovery requires a connection |
| GitHub, Jira, Spec Kit | Development process | Learning curve |

## 5. Product Features

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

### 5.1 Authentication & Security

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

This feature lets users register, log in, log out, and recover a forgotten password through an expiring reset link. Protected online requests use validated JWT access tokens, while offline access is limited to locally stored data of the previously signed-in user. Access control separates personal records, group membership, and administrator functions because the app stores sensitive financial information. Regular Users and Administrators benefit from a consistent account lifecycle and protection against unauthorized access.

### 5.2 Profile Management

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Users can edit their name and avatar, change their password, select Vietnamese or English, and choose VND or USD as their reporting currency. Reports show converted totals with exchange-rate details while retaining the original currency on each financial entry. These preferences make the app understandable to local users and let group members recognize one another. Regular Users benefit from consistent identity and financial presentation across personal and group screens.

### 5.3 Wallet & Personal Account Management

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Users create manual cash, bank-account, and credit-card wallets, each with a recorded currency and opening balance. The dashboard shows individual balances and a combined estimate in the selected reporting currency using an identified reference rate. Cached balances remain readable offline, and pending personal transactions update the local view until synchronization. This helps Regular Users understand their recorded assets without connecting to a real bank account.

### 5.4 Personal Transaction Tracking

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Users record income and expenses with amount, category, wallet, note, date, and an optional receipt image. They can create, edit, or delete personal transactions offline, with changes and receipt uploads queued until reconnection. Stable record identifiers prevent duplicates, while conflicting edits remain visible for resolution rather than being silently overwritten. Regular Users benefit from reliable capture at the time of payment and a transaction history that supports budgets, reports, and AI analysis.

### 5.5 Group Management & Shared Wallets

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Users create groups for shared activities, invite members by link or code, and use Owner and Member roles to control group access. A group records its currency, shared-wallet contributions, expenses, and member balances separately from private wallets. Group changes require a connection so all members work from the same confirmed ledger, while previously synchronized information remains readable offline. Friends, roommates, and teammates benefit from a transparent record of who contributed, who spent, and who should be reimbursed.

### 5.6 Smart Debt Split & Settlement

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Shared expenses are divided equally, by percentage, or by exact amount after any required conversion into the group's currency. The app shows confirmed debts and suggests fewer repayment transfers while preserving each member's net balance. A debtor records an external payment claim, and the recipient confirms it before the debt is reduced; partial payments and rejected claims remain traceable. Refunds and corrections reference the original expense and adjust member balances without deleting payment history. Group members benefit from consistent calculations, reminders, and an auditable settlement process.

### 5.7 Budget Planning & Alerting

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Users set weekly or monthly limits for personal spending categories in their selected budget currency. Spending includes their own group-expense shares once, using the relevant recorded conversion rates. Visual warnings appear when a category reaches 80% and 100% of its limit, including for locally recorded offline expenses. Regular Users benefit from timely feedback instead of discovering overspending only at the end of the month.

### 5.8 Financial Reporting & Analytics

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

The app presents personal income and spending in pie and bar charts and provides group summaries for trips or recurring shared costs. Reports separate consumption from transfers, contributions, and debt settlements, and display the reporting currency and exchange-rate basis. Authorized users can export PDF or Excel reports online, while cached summaries remain readable offline with their last-sync time. Individuals benefit from understanding spending patterns, and groups benefit from clear records they can review together.

### 5.9 Savings Goal Management

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Users create savings goals with a target amount, currency, deadline, and recorded contributions or withdrawals. The app shows progress and the remaining amount, using recorded conversion rates when a contribution has another currency. A savings allocation moves recorded funds toward a goal without creating new income or counting the same expense twice. Students and young professionals benefit from a measurable saving plan and can view previously synchronized progress offline.

### 5.10 System Administration (Admin Panel)

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Administrators use an online web panel to list users, lock or unlock accounts, maintain default categories and reference exchange rates, and view aggregate system statistics. Changes to rates retain their source, effective time, and author so converted amounts can be explained. Administrative access does not grant routine access to private transaction histories or permission to confirm another member's settlement. Administrators benefit from consistent maintenance tools, and Regular Users benefit from traceable reference data and clear privacy boundaries.

### 5.11 AI Financial Advisor

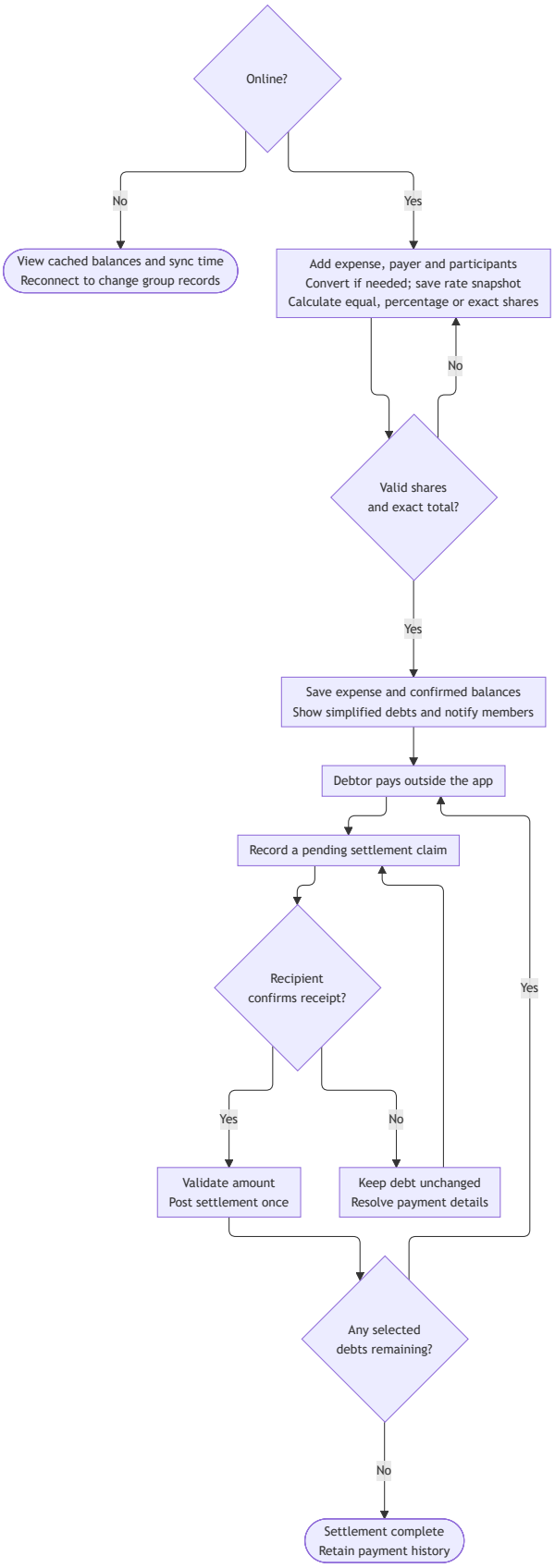
> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

At the start of a month or on request, the AI analyzes the user's personal spending and attributable group shares, excluding duplicate entries and using recorded currency conversions. It compares available history with budgets, identifies unusual category increases, and suggests a practical allocation for the next month and savings goals. New analysis requires a connection, explains limited history when applicable, and leaves existing finance functions available if the provider fails. Regular Users can review previously saved advice offline and choose whether to apply a suggestion, with no automatic change to budgets or financial records.

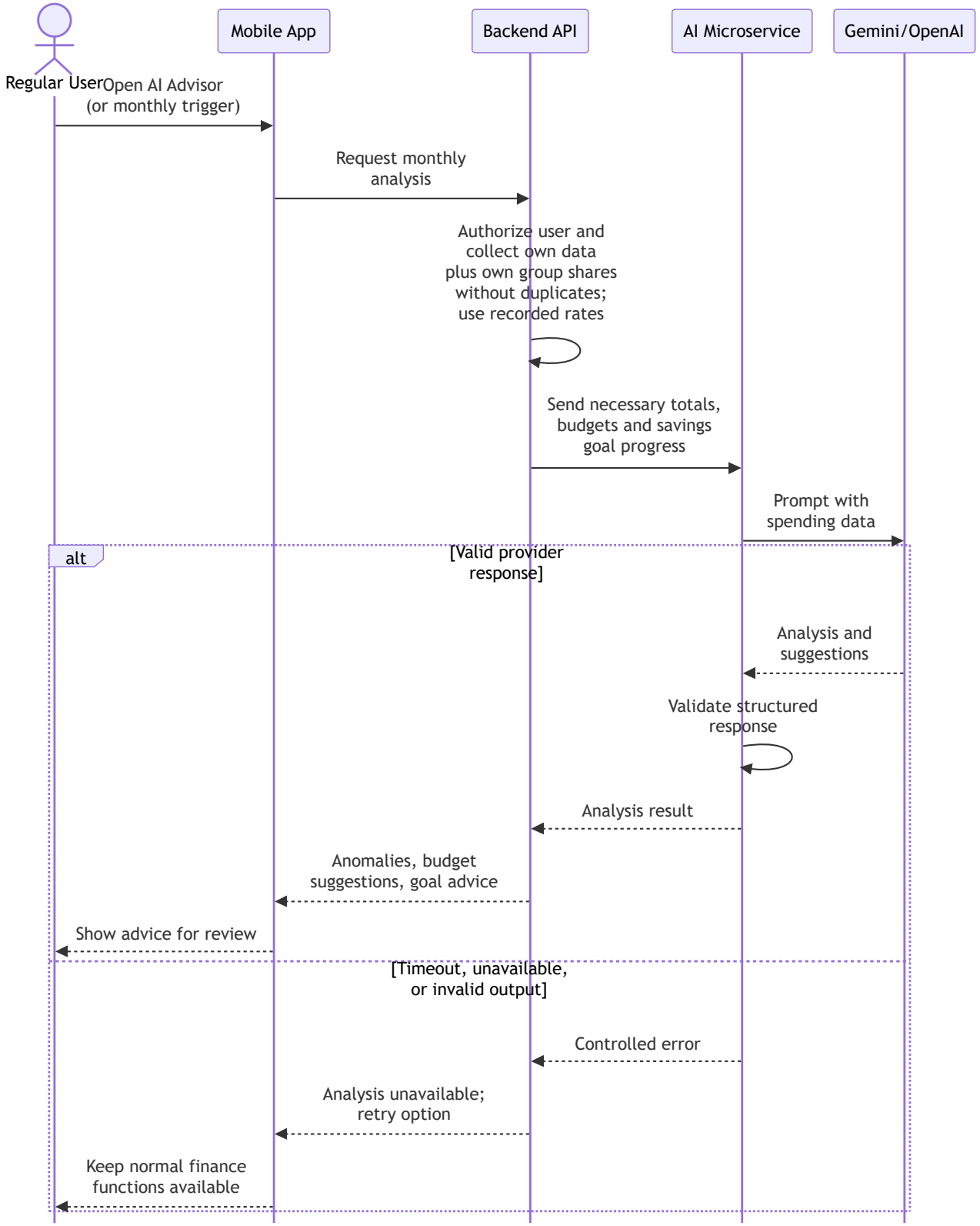
### 5.12 Key Workflow Diagrams

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

## Workflow 1 — Record a shared expense and settle debts



**Workflow 2 — Monthly AI Financial Advisor**




Workflow 1 requires an online ledger and confirmation by the payment recipient before a debt changes. Workflow 2 operates on the signed-in user's synchronized data and recorded conversion rates; offline users can read cached advice and personal records. Both workflows preserve the separation between external payments and in-app bookkeeping.

## 6. Non-Functional Requirements

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

The following requirements define acceptance criteria for implementation and final testing. Performance tests use a Linux deployment with at least **2 vCPUs and 4 GiB RAM**, a seeded database of **100 users with 100 personal transactions each**, and group test cases of up to **20 members and 500 expenses per group**. Mobile timing tests use an Android device with **4 GiB RAM**, Android 10 or later, and a stable connection of at least **10 Mbps** with round-trip latency no greater than **100 ms**. Minimum-platform compatibility is checked separately on Android 8.0; every test report records the actual hardware, versions, dataset, and build revision.

### 6.1 Performance

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

| ID | Requirement |
|---|---|
| NFR-P1 | At least 95% of non-AI API requests complete in **less than 500 ms** during the NFR-P3 load test. |
| NFR-P2 | The dashboard and first page of up to 50 transactions appear in **less than 2 seconds** in at least **19 of 20** warm-load trials on the specified mobile test environment. Cached offline lists meet the same target. |
| NFR-P3 | Support **100 concurrent authenticated users for 10 minutes**, each making one non-AI request every two seconds, with an unexpected-error rate **below 1%**. The request mix is 70% reads and 30% personal transaction writes; expected validation and authorization rejections are reported separately. |
| NFR-P4 | An online AI request produces advice or a controlled failure response within **30 seconds**. The interface shows progress and a retry option; successful and failed analysis counts are reported separately. |
| NFR-P5 | Recalculate group balances and simplified repayment suggestions for **20 members and 500 expenses in less than 2 seconds**, preserving each member's net position. |
| NFR-P6 | With the app open, valid authorization, and the specified connection, synchronize **100 queued personal transaction changes within 30 seconds**, excluding receipt uploads and edits awaiting conflict resolution. |

### 6.2 Security

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

| ID | Requirement |
|---|---|
| NFR-S1 | Store passwords only as salted BCrypt hashes; no plaintext or recoverable password is stored or logged. |

| ID | Requirement |
|---|---|
| NFR-S2 | Use **HTTPS with TLS 1.2 or later** for deployed client/API and external-service communication. Local development uses isolated test data. |
| NFR-S3 | Access tokens expire after **60 minutes**; refresh sessions expire after **7 days**. Protected requests reject invalid or expired tokens. Logout removes local tokens and private cached data, revokes the refresh session, and invalidates its active access token on the server. Synchronization after session expiry requires reauthentication. |
| NFR-S4 | All private-record, group-membership, administrator-role, and settlement-recipient authorization checks pass the agreed negative-test suite. A Group Owner cannot confirm a payment intended for another member. |
| NFR-S5 | Keep API keys, passwords, signing secrets, and real environment files outside tracked source and submission files. Application logs and AI requests must omit credentials and access tokens. |
| NFR-S6 | Validate amounts, currencies, split totals, uploaded file types, and user input on the server. Use parameterized database access and safe UI rendering; all invalid-input and script-injection cases in the security suite must pass. |
| NFR-S7 | Send the AI provider only the signed-in user's necessary category totals, budget limits, own group shares, and goal progress. Exclude other members' private records, names, emails, phone numbers, raw receipts, and authentication data. |
| NFR-S8 | Encrypt private offline records and receipt files at rest with keys protected by the Android system keystore. Cached data belongs to one signed-in account, requires device/app unlock to access, and is cleared on explicit logout or account removal from the device. |
| NFR-S9 | Password-reset links expire after **15 minutes**, can be used only once, and do not expose whether an unknown email address has an account. |

### 6.3 Reliability, Offline Operation, and Financial Correctness

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

| ID | Requirement |
|---|---|
| NFR-R1 | The demo environment passes **at least 95%** of one-minute health probes during a **two-hour rehearsal** before the demo. |
| NFR-R2 | Back up hosted project data **daily**, retain the latest **7 daily backups**, and demonstrate restoration to a separate test database within **30 minutes** before the final demo. The QA/DevOps Lead owns backup and restore verification. |
| NFR-R3 | Use exact decimal arithmetic with whole-dong VND and cent-precision USD. In every split/refund test, allocated shares equal the recorded amount exactly, and every group debt/fund journal balances to zero. Stable remainder allocation produces the same result on repeated calculation. |

| ID | Requirement |
|---|---|
| NFR-R4 | During an AI-provider outage or invalid response, personal tracking, group bookkeeping, and cached advice remain available within their normal connectivity limits; no financial record or budget is changed by a failed AI request. |
| NFR-R5 | After initial synchronization, users can create, edit, and delete personal transactions for cached wallets with **no network connection**. A **24-hour offline test with 100 local changes**, including app restart, loses no acknowledged local change and retains pending status. |
| NFR-R6 | Replaying each queued change **three times** creates no duplicate financial effect. Conflicting edits from two devices are detected, both versions are retained, and the user resolves the conflict before that record is marked synchronized. |
| NFR-R7 | A payment claim remains pending until the intended recipient confirms. Rejection, repeated confirmation, server failure, and attempts to exceed the outstanding debt produce **no incorrect balance change** in the settlement test suite; dependent ledger updates commit together or roll back together. |
| NFR-R8 | Every converted entry stores its original amount/currency and applied rate/source/time. Changing the reference rate or reporting preference does not alter existing group shares, original amounts, or confirmed payments. Missing rates never produce an invented converted total. |

### 6.4 Usability and Accessibility

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

| ID | Requirement |
|---|---|
| NFR-U1 | At least **4 of 5** first-time test participants record their first personal transaction in **3 minutes or less** without assistance, starting from an authenticated account with a wallet. |
| NFR-U2 | At least **4 of 5** participants save a basic expense in **30 seconds or less** from the add-transaction screen, excluding receipt capture/upload. Use the same task script online and offline and record results separately. |
| NFR-U3 | All primary screens use consistent labels and components; text supports Android font scaling to **150%** without hiding required actions, and primary touch targets are at least **48 dp**. |
| NFR-U4 | Provide Vietnamese and English labels for all primary flows, VND/USD formatting, original and converted amounts, and the applied rate's effective date. Amounts are always labelled with their currency. |
| NFR-U5 | Offline, pending synchronization, conflict, and pending/confirmed settlement states are distinguishable using text or icons as well as color. Shared-data views show their last successful synchronization time. |

| ID | Requirement |
|---|---|
| NFR-U6 | Budget warnings are updated when a relevant local or synchronized expense causes spending to cross **80% or 100%** of its limit; offline warnings identify that group data may be stale. |

### 6.5 Platform and Compatibility

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

| ID | Requirement |
|---|---|
| NFR-C1 | The Kotlin/Jetpack Compose mobile app installs and supports the core personal and group flows on **Android 8.0, API 26, or later**. Compatibility is checked on API 26 and an Android 10-or-later device/emulator. |
| NFR-C2 | The online admin panel supports the **two most recent stable desktop versions** of Chrome, Edge, and Firefox available at final acceptance; record the tested version numbers. |
| NFR-C3 | The server targets **Java 21 and PostgreSQL 16**. Backend, database, AI service, and admin deployment are reproducible on a Docker-capable Linux host using documented Docker Compose commands and external configuration. |
| NFR-C4 | Receipt capture works without a connection and uploads later. Online recovery, new AI requests, group writes, and settlement confirmation clearly indicate their network requirement; viewing cached data remains available. |

### 6.6 Maintainability and Quality

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

| ID | Requirement |
|---|---|
| NFR-M1 | Automated tests achieve **at least 70% line coverage** of backend services implementing splitting, settlement, refunds, currency conversion, and budget calculations. Coverage is measured against the reviewed build. |
| NFR-M2 | Follow Java naming conventions, the Kotlin Style Guide, PEP 8, and the agreed React conventions. Another member reviews each change before merge; review findings are resolved or explicitly accepted. |
| NFR-M3 | Document all public REST endpoints, authorization rules, request/response schemas, and error responses in OpenAPI. Document synchronization identifiers and conflict responses. |
| NFR-M4 | Apply server database changes through versioned Flyway migrations and verify both a clean setup and upgrade from the previous test build. |

| ID | Requirement |
|---|---|
| NFR-M5 | Use Conventional Commits and retain traceability from a Jira task to its implementation/document output and review record. |
| NFR-M6 | Automated financial tests include equal/percentage/exact splits, rounding remainders, partial/rejected settlements, refunds after settlement, duplicate retries, offline conflicts, and VND/USD conversion using fixed test rates. |

### 6.7 Standards and Verification

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

The API uses REST with JSON and OpenAPI documentation. Security review uses the OWASP Top 10 as a checklist, alongside the explicit authorization and data-protection requirements above. Money calculations follow the stated VND/USD precision, half-up conversion, immutable rate-snapshot, and exact-sum rules. Development follows Scrum, peer review, Conventional Commits, and versioned database migrations.

| Requirement group | Verification method | Acceptance evidence produced during testing |
|---|---|---|
| Performance, P1–P6 | Defined mixed API load, mobile timing trials, AI timeout simulation, debt benchmark, and queued-change sync timing | Load configuration, timing/error results, actual environment, and tested revision |
| Security, S1–S9 | Password/token/reset tests, role and ownership tests, invalid-input cases, secret checks, AI-payload inspection, and offline-cache/logout checks | Security checklist, test results, reviewed payload samples, and protected-data checks |
| Reliability, R1–R8 | Rehearsal health probes, backup/restore, 24-hour offline/restart exercise, repeated retries, concurrent edits, ledger rollback, and historical-rate tests | Health log, restore record, synchronization/conflict results, and financial invariant test results |
| Usability, U1–U6 | Timed online/offline tasks with five first-time participants, language review, font/touch-target checks, and budget-threshold scenarios | Anonymized task timings, screen captures, accessibility checks, and alert results |

# Part C — Spec Kit Initialization

> *Performed by:* `Phạm Đinh Tiểu Long` | *Reviewed by:* `Nguyễn Lê Đức Nhật` | *Edited by:* `Phạm Đinh Tiểu Long`

## 1. Repository and Spec Kit Setup

> *Performed by:* `Phạm Định Tiểu Long` | *Reviewed by:* `Nguyễn Lê Đức Nhật` | *Edited by:* `Phạm Định Tiểu Long`

| Item | Value |
|---|---|
| GitHub repository | pdtLong2929/FinSync, recorded in PA1; `verified repository access for TA` |
| Spec Kit location | `/src` |
| Initialization status | Historical initialization snapshot at commit `b98938f`; this does not describe the current `speckit` branch layout |
| AI integration at initialization | GitHub Copilot (`copilot`), as recorded in the files at `b98938f` |
| Spec Kit version at initialization | `1.2.1`, as recorded in `src/.specify/init-options.json` and `src/.specify/integration.json` at `b98938f` |
| Initial setup date reported by the team | `19/10/2026`; this is the reported setup date, not by itself evidence of constitution ratification or full S2-12 acceptance |
| Script environment at initialization | PowerShell (`ps`), with generated scripts under `src/.specify/scripts/powershell/` at `b98938f` |
| Setup performer / reviewer | `Phạm Định Tiểu Long` / `Nguyễn Lê Đức Nhật` |
| Earlier setup commit | `6c5cd20` — `add: new speckit`; contains an empty `src/constitution.md` |
| Initialization artifact evidence | `b98938f` — `chore: initialize Spec Kit in src`, committed on `19/10/2026` |

**Configuration recorded at the initialization commit**

The following configuration is recorded in `src/.specify/init-options.json` at commit `b98938f`. It describes that commit only, not the current branch configuration:

```
{
  "ai": "copilot",
  "ai_skills": true,
  "allow_symlinks": false,
  "feature_numbering": "sequential",
  "here": true,
  "integration": "copilot",
  "script": "ps",
  "speckit_version": "1.2.1"
}
```

The companion `src/.specify/integration.json` at `b98938f` records Copilot as the installed and default integration, with PowerShell scripts. At that commit, the initialization artifacts were under `src/.specify/` and `src/.github/skills/`; the constitution path was `src/.specify/memory/constitution.md`.

## 2. Submitted Artifacts

> *Performed by:* `Phạm Định Tiểu Long` | *Reviewed by:* `Nguyễn Lê Đức Nhật` | *Edited by:* `Phạm Định Tiểu Long`

| **Artifact present in initialization commit** | **Repository path at `b98938f`** | **Evidence** |
|---|---|---|
| speckit-analyze instructions | `src/.github/skills/speckit-analyze/SKILL.md` | `b98938f` |
| speckit-checklist instructions | `src/.github/skills/speckit-checklist/SKILL.md` | `b98938f` |
| speckit-clarify instructions | `src/.github/skills/speckit-clarify/SKILL.md` | `b98938f` |
| speckit-constitution instructions | `src/.github/skills/speckit-constitution/SKILL.md` | `b98938f` |
| speckit-converge instructions | `src/.github/skills/speckit-converge/SKILL.md` | `b98938f` |
| speckit-implement instructions | `src/.github/skills/speckit-implement/SKILL.md` | `b98938f` |
| speckit-plan instructions | `src/.github/skills/speckit-plan/SKILL.md` | `b98938f` |
| speckit-specify instructions | `src/.github/skills/speckit-specify/SKILL.md` | `b98938f` |
| speckit-tasks instructions | `src/.github/skills/speckit-tasks/SKILL.md` | `b98938f` |
| speckit-taskstoissues instructions | `src/.github/skills/speckit-taskstoissues/SKILL.md` | `b98938f` |
| Constitution file (template with placeholders at this commit) | `src/.specify/memory/constitution.md` | `b98938f` |
| checklist-template.md | `src/.specify/templates/checklist-template.md` | `b98938f` |
| constitution-template.md | `src/.specify/templates/constitution-template.md` | `b98938f` |
| plan-template.md | `src/.specify/templates/plan-template.md` | `b98938f` |
| spec-template.md | `src/.specify/templates/spec-template.md` | `b98938f` |

| Artifact present in initialization commit | Repository path at `b98938f` | Evidence |
|---|---|---|
| tasks-template.md | `src/.specify/templates/tasks-template.md` | `b98938f` |

At commit `b98938f`, the initialization inventory consists of 16 Markdown files: ten workflow instruction files, five templates, and the constitution file. The constitution in this historical snapshot is a template with placeholders, not a completed project constitution. This inventory documents the initialization commit only; it is not an inventory of the current `speckit` branch. The earlier `6c5cd20` commit contains an empty `src/constitution.md` and is separate from this initialization output.

**Completed project artifact:** `.specify/memory/constitution.md` — FinSync Constitution, Version `1.0.1`, committed in `9cb4ef4`. This is the completed constitution in the current repository-root Spec Kit layout; it is distinct from the placeholder template listed in the historical `b98938f` inventory above.

## 3. Constitution Summary

> *Performed by:* `Phạm Đinh Tiểu Long` | *Reviewed by:* `Nguyễn Lê Đức Nhật` | *Edited by:* `Phạm Đinh Tiểu Long`

The current constitution is located at `.specify/memory/constitution.md` in the repository-root Spec Kit layout. It defines six principles:

1. **Non-Custodial & Manual Settlement:** FinSync is a bookkeeping and advisory system; real payments happen outside the application, and the system must not hold funds or initiate bank transfers.
2. **Mobile Client Architecture:** Android uses Kotlin, Jetpack Compose, Coroutines, and Retrofit; the specified cross-platform option uses Flutter with BLoC or Riverpod and Dio.
3. **Backend Service Architecture:** The API uses Spring Boot 4.1.1 and Java 21 with layered REST architecture, Spring Security/JWT, Flyway migrations, and SpringDoc OpenAPI.
4. **AI & Data Parsing Microservice Isolation:** AI advisory and notification parsing are assigned to a separate Python/FastAPI service integrating with Gemini and/or OpenAI.
5. **Web Admin Architecture:** The administrative portal uses React.js for platform monitoring and user-account governance.
6. **Managed Persistence & Relational Schema:** PostgreSQL 16 is the primary database; schema changes must use versioned Flyway migrations.

The constitution also states technology-stack, security, development-quality, and governance requirements. Its recorded metadata is Version `1.0.1`, Ratified `2026-10-20`, and Last Amended `2026-10-22`. The earlier snapshot at commit `b98938f` under `src/.specify/memory/constitution.md` was still a template with placeholders; it is historical evidence of initialization, not the current constitution.

## 4. Spec Kit Training Evidence

> *Performed by:* `Phạm Đinh Tiểu Long` | *Reviewed by:* `Nguyễn Lê Đức Nhật` | *Edited by:* `Phạm Đinh Tiểu Long`

**Shared training evidence:** Spec Kit training records for all five members.

**Tutorial reference:** GitHub Spec Kit tutorial. The table records the team's reported completion dates, individual learning summaries and reviewers. The evidence for all five members is stored in the shared folder above.

| Member | Completed / Date | Individual Learning Summary | Reviewer / Date |
|---|---|---|---|
| Phạm Định Tiểu Long | 19/10/2026 | Completed tutorials #1-#7. Practiced end-to-end Spec Kit workflow and repo init. Key lesson: strict rules in constitution.md effectively prevent AI architectural drift. | Hòa / 20/10/2026 |
| Nguyễn Phú Đạt | 19/10/2026 | Learned how the `/clarify` command identifies edge cases before planning and reduces rework. | Long / 20/10/2026 |
| Nguyễn Lê Đức Nhật | 20/10/2026 | Gained hands-on experience turning user ideas into concrete specs using /specify; learned that thorough acceptance criteria make the final task checklist much easier to execute. | Đạt / 20/10/2026 |
| Vương Đắc Gia Khiêm | 20/10/2026 | Learned to use the interactive `/clarify` command to refine requirements before code generation and align them with the project constitution. | Nhật / 20/10/2026 |
| Ngô Thái Hòa | 20/10/2026 | Realized that spec-driven development eliminates guesswork and trial-and-error by enforcing clear structure right from workspace initialization. | Khiêm / 20/10/2026 |

All members have completed the entire tutorial series (Parts 1 to 7) to gain an end-to-end understanding of GitHub Spec Kit. To thoroughly assess individual comprehension and ensure mastery of each stage, specific modules were assigned to each member for in-depth verification:

- **Long:** Part 2 – Constitution File (`constitution.md` principles & governance)
- **Nhật:** Part 3 – The `/specify` Command (Feature specification & PRD generation)
- **Hòa:** Part 4 – The `/clarify` Command (Resolving ambiguities & handling edge cases)
- **Khiêm:** Part 5 – The `/plan` Command (Technical architecture & implementation planning)
- **Đạt:** Part 6 – The `/tasks` Command (Task breakdown & execution tracking)

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# Part D — AI Usage Report and Weekly Report

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Nguyen Le Duc Nhat`

## 1. AI Usage Report

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Nguyen Phu Dat` | *Edited by:* `Nguyen Le Duc Nhat`

**Declaration:** All five members have supplied individual AI usage entries for PA2. The submitted records identify Codex, ChatGPT, Claude, and Gemini. Nhật declares Codex-only use; Long lists Claude and Gemini; Đạt and Khiêm list ChatGPT; Hòa lists Gemini, Claude, and ChatGPT. Their model labels, dates, prompts, purposes, and contribution descriptions are retained in the Appendix.

**Reporting scope:** AI usage through the 22 October 2026 Sprint 2 Review; 65 prompt entries from all five members, dated 10-22 October 2026. Tool metadata and dates follow member declarations; requested assistance, student contributions, and completed validation are distinguished in the Appendix.

| Member | Entries | Tools / models as declared | Entry dates | Main purposes |
|---|---|---|---|---|
| Nguyễn Lê Đức Nhật | 14 | Codex (`gpt-6.1-sol`) | 10–21 October | Report structure, planning, business requirements, workflows, NFRs, and Spec Kit learning |
| Phạm Định Tiểu Long | 15 | Claude (`3-opus`); Gemini (`1.5-pro`) | 10–22 October | Architecture, database, APIs, backend examples, tests, and reflection |
| Nguyễn Phú Đạt | 12 | ChatGPT (`gpt-6.1-sol`) | 10–21 October | Windows setup, Copilot integration, artifact verification, training, and Part C documentation |
| Vương Đắc Gia Khiêm | 13 | ChatGPT (GPT-6) | 14–20 October | Test Strategy, testing concepts, CI/CD, quality gates, and Spec Kit learning |
| Ngô Thái Hòa | 11 | Gemini (`gemini-2.5-pro`); Claude (Sonnet 5.5); ChatGPT (GPT-6) | 10–21 October | macOS environment setup, architecture, Spec Kit workflows, stalled execution, and quota troubleshooting |

| Purpose | AI-generated assistance | Affected sections | Human input and validation responsibility |
|---|---|---|---|
| Report organization and English wording | Proposed section structure, refined descriptions, and prepared document tables | Parts A–D and the Appendix | Nhật supplied the assignment, existing report, and project context; section authors check the resulting meaning and assignment coverage |
| Requirements and business rules | Drafted offline boundaries, group accounting, refunds, settlement confirmation, and VND/USD conversion rules | Part B, Sections 3–5 | Nhật confirmed offline support, the unchanged PA1 survey, adoption of the accounting rules, and currency conversion; assigned reviewers check technical accuracy |

| Purpose | AI-generated assistance | Affected sections | Human input and validation responsibility |
|---|---|---|---|
| Sprint planning and workload | Proposed assignments, dates, dependencies, estimates, capacity allocation, and builds | Part A, Sections 4.2–4.5 | The team supplied roles and the sprint window; Long coordinates the plan and reviewers check the assigned outputs |
| Workflows and quality requirements | Drafted Mermaid workflows, measurable NFR targets, and verification methods | Part B, Sections 5.12 and 6 | Authors check workflow logic, accounting examples, and whether the proposed criteria can be tested |
| Weekly reporting and evidence | Drafted role-based meeting records, captions, and the AI disclosure | Part D and the Appendix | Nhật confirmed past attendance and the corrected meeting dates and completion of all assigned sprint work at the 22 October Review, as recorded in D.2 |
| Document checking and revision | Located unfinished fields, corrected caption CSS and Mermaid syntax, reduced Jira evidence to one board image, and aligned reviewers and packaging dates | Report and PDF stylesheet | Nhật requested these corrections; document checks compare the final wording, local image paths, task baseline, and supplied evidence |
| Test Strategy and Spec Kit learning | Feedback on Test Strategy structure, a requested draft of TEST_STRATEGY.md, and explanations of testing methods, CI/CD, quality gates, and Spec Kit workflows | Test Strategy document; Part C; Appendix, Section 4 | Khiêm supplied the proposed structure and learning questions. His 13 entries describe the requested assistance and completed student validation. |
| Architecture and backend assistance | Proposed package/domain boundaries, schema and API drafts, diagrams, code examples, configuration, tests, and a reflection draft | Backend design/work; related Part B topics; Appendix, Section 2 | Long supplied the requests and reported adaptations or checks in his 15 entries; related artifacts document implementation and performance work. |

| Purpose | AI-generated assistance | Affected sections | Human input and validation responsibility |
|---|---|---|---|
| Spec Kit setup and training assistance | Suggested Windows setup steps, Copilot initialization, artifact/Git checks, and training-summary wording | Part C; Appendix, Section 3 | Đạt supplied environment questions and evidence-related requests; his 12 entries describe the completed local checks and training comparison. |
| Environment and AI workflow troubleshooting | Suggested macOS CLI/PATH fixes, collaboration and architecture guidance, workflow examples, and diagnostics for stalled agents or quotas | Part C; development environment and AI workflow; Appendix, Section 5 | Hòa supplied commands, symptoms, and error details; his 11 entries include reported configuration changes and troubleshooting purposes. |

**Student contribution and validation**

Nhật selected the source documents, supplied the project decisions, identified inconsistencies, and directed revisions through the recorded prompts. AI supplied substantial draft wording and planning proposals. Document checks cover section structure, assignee/reviewer separation, date dependencies, PA1 survey consistency, image references, and the distinction between requirements and measured results. Assigned team reviewers remain responsible for recording their own review decisions against the corresponding tasks.

**Supporting records**

The Appendix contains individual records for all five members: Nhật's documentation and requirements assistance; Long's architecture, backend, database, and API assistance; Đạt's Spec Kit setup and training assistance; Khiêm's testing and Spec Kit learning assistance; and Hòa's environment troubleshooting, architecture, and Spec Kit workflow assistance. Nhật's English prompt text is translated from the supplied Vietnamese entries. The team confirms that all listed AI usage occurred on or before 22 October 2026. Entry dates follow member declarations and the summary ranges match the Appendix.

Hòa is responsible for consolidating the group disclosure under S2-24, with Long as reviewer.

## 2. Weekly Report

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

**Project:** FinSync — Group 04.
**Sprint:** Sprint 2 / PA2.
**Period:** 2026-10-10 to 2026-10-25.

**Sprint goal:** Prepare an initial Project Plan and Vision Document, initialize Spec Kit under /src, collect individual training evidence, and produce a reproducible skeleton build that supports later

implementation.

The structure follows `WeeklyReport`: one Sprint Planning meeting, two Weekly Scrum meetings, and one Sprint Review/Retrospective. Both Scrum records are kept in this document. Long coordinates the meetings and minutes; all five members provide their own progress updates.

| Meeting | Confirmed date / scheduled time | Task | Minutes owner / reviewer |
|---|---|---|---|
| Sprint Planning | 2026-10-10, 09:00–10:00, before sprint work begins | S2-01 | Long / Hòa |
| Weekly Scrum #1 | 2026-10-14, 19:30–20:15 | S2-21 | Long / Hòa |
| Weekly Scrum #2 | 2026-10-18, 19:30–20:15 | S2-22 | Long / Hòa |
| Sprint Review & Retrospective | 2026-10-22, 19:30–20:30 | S2-23 | Long / Khiêm |

The four sprint meetings took place on 10, 14, 18, and 22 October. Planning is the PA1-to-PA2 transition checkpoint on 10 October before implementation begins. Meetings use Google Meet, with follow-up coordination through Discord/Zalo.

### 2.1 Sprint Planning Meeting

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

**Date / scheduled slot:** 2026-10-10, 09:00–10:00.
**Facilitator / note taker:** Long.
**Members present:** Phạm Định Tiểu Long, Nguyễn Phú Đạt, Nguyễn Lê Đức Nhật, Vương Đắc Gia Khiêm, and Ngô Thái Hòa.
**Members absent:** None; full attendance confirmed by the group.

**Prioritized scope and allocation**

| Priority | Work package | Assigned owner(s), each for their own tasks | Expected output |
|---|---|---|---|
| 1 | Project Plan, schedule, risk management, and coordination | Long; Nhật for Introduction/Overview | Initial plan covering five sprints, task/reviewer assignments, risks, and builds |
| 1 | Vision, positioning, user needs, and product boundaries | Hòa for positioning/features; Đạt for user research; Nhật for product perspective | Refined descriptions of all ten functional groups and AI Advisor; two Mermaid workflows |

| Priority | Work package | Assigned owner(s), each for their own tasks | Expected output |
|---|---|---|---|
| 1 | Spec Kit setup and training evidence | Long for setup; Khiêm for evidence collection; every member for individual training | Customized constitution, generated Markdown inventory, and five individual evidence records |
| 1 | Quality targets and PA2 evidence | Khiêm for NFRs/evidence; Hòa for the AI log; Long for minutes | Measurable NFRs, AI Usage Notes, four meeting records, and Jira/Git evidence |
| 2 | Mobile/backend/AI skeletons and Build 0 | Đạt for mobile; Nhật for backend; Hòa for AI; Khiêm for integration | Reproducible startup instructions and documented smoke-test results |

Each task has one assignee and a different reviewer, as listed in Section A.4.3. The baseline allocates 99 team hours against approximately 114 available hours, including separate review/attendance budgets and a correction buffer.

**Planning baseline**

- Retain the Android Kotlin/Jetpack Compose baseline and the Spring Boot/PostgreSQL backend described in PA1.
- Keep personal records private and group records visible only to authorized members. Record real payments outside the app.
- Complete training by 12 October, Spec Kit setup by 15 October, feature drafts by 17 October, and skeleton work by 20 October.
- Produce Build 0 by 22 October. The original non-meeting baseline allowed review and packaging through 25 October; the team subsequently confirmed completion of all assigned work at the 22 October Review.
- Track all writing, learning, coding, review, and evidence tasks in Jira; update actual status and dates from real work.

**Follow-up actions:** ACT-01 and ACT-02 in Section 2.7.

**Summary:** The sprint baseline prioritizes mandatory PA2 documents and training while reserving time for early technical integration. Product descriptions guide workflows and quality targets; component skeletons precede Build 0. The current scope also incorporates the agreed product decisions on offline support, recipient-confirmed settlement, and VND/USD conversion, as consolidated in Part B.

### 2.2 Weekly Scrum Meeting #1 — 14 October 2026

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

**Date / scheduled slot:** 2026-10-14, 19:30–20:15.
**Facilitator / note taker:** Phạm Định Tiểu Long.
**Members present:** Phạm Định Tiểu Long, Nguyễn Phú Đạt, Nguyễn Lê Đức Nhật, Vương Đắc Gia Khiêm, and Ngô Thái Hòa.
**Members absent:** None; full attendance confirmed by the group.

**Meeting focus:** Review the first documentation drafts, identify work behind the initial schedule, and prepare the mobile/backend/AI integration checkpoint.

**Progress:** Initial document drafts were available. Spec Kit initialization and individual training were still in progress.

| Member | What have I done since last week? | What will I do until next week? | Issues / problems / obstacles |
|---|---|---|---|
| Long | Initial organization, risk, schedule, and build-plan drafts are available in the shared report under S2-03/S2-04. | Complete Spec Kit setup, the constitution, and artifact inventory (S2-12); coordinate document reviews and training recovery before the next Scrum. | The training/setup workstream is behind its original early-sprint milestones; draft documents alone do not establish readiness for integration. |
| Đạt | User/stakeholder descriptions and the PA1 survey baseline are available for the vision draft under S2-06. | Complete the mobile skeleton (S2-18), finalize two workflows (S2-09), and review the integrated Build 0. | The mobile client needs agreed request/response fields, connectivity states, and currency/accounting behavior before integration. |
| Nhật | Introduction and product-perspective drafts are available. | Complete the backend/database/API skeleton (S2-19), review the Spec Kit setup, and align the backend/AI contract. | Exact money representation, duplicate prevention, and group/personal boundaries require consistent treatment across components; successful integration is not yet evidenced. |
| Khiêm | Repository preparation and the Spec Kit/training workstream have been identified for technical follow-up; completed initialization/training evidence is not yet available. | Support Long on environment checks for S2-12, collect each member's training evidence, define the Build 0 checklist, and prepare NFR verification methods (S2-10). | Spec Kit/training remain incomplete against the 12/15 October baseline targets and need a recovery checkpoint. |

| Member | What have I done since last week? | What will I do until next week? | Issues / problems / obstacles |
|---|---|---|---|
| Hòa | Vision introduction/positioning and product-feature descriptions are available as initial draft material under S2-05/S2-08. | Refine the features with peer feedback, complete the AI skeleton (S2-20), and consolidate AI-use evidence (S2-24). | AI-provider timeout, quota, and invalid-output handling remain integration risks to test; per-member usage histories must be collected. |

**Follow-up reconciliation — 19 October 2026**

The 19 October follow-up set recovery targets for training, Spec Kit setup, and document review.

| Follow-up | Task / expected output | Owner | Reviewer | Recovery target |
|---|---|---|---|---|
| SC1-01 | Complete Long's training and retain an individual learning/evidence record (S2-13) | Long | Hòa | 2026-10-20 |
| SC1-02 | Complete Đạt's training and retain an individual learning/evidence record (S2-14) | Đạt | Long | 2026-10-20 |
| SC1-03 | Complete Nhật's training and retain an individual learning/evidence record (S2-15) | Nhật | Đạt | 2026-10-20 |
| SC1-04 | Complete Khiêm's training and retain an individual learning/evidence record (S2-16) | Khiêm | Nhật | 2026-10-20 |
| SC1-05 | Complete Hòa's training and retain an individual learning/evidence record (S2-17) | Hòa | Khiêm | 2026-10-20 |
| SC1-06 | Complete the project constitution and verify the generated-file inventory under /src after initial setup (S2-12) | Long | Nhật | 2026-10-21 |
| SC1-07 | Coordinate outstanding document reviews and record actual progress/recovery dates in Jira (ACT-03/ACT-04) | Long | Hòa | 2026-10-21 |

**Summary:** Initial plan and vision drafts provide a basis for review, but Spec Kit/training are the confirmed schedule gap. The recovery plan prioritizes individual evidence and reproducible setup before the 22 October integration checkpoint. Scope is kept stable so the remaining days can be used for technical checks, review, and submission evidence.

### 2.3 Weekly Scrum Meeting #2 — 18 October 2026

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

**Date / scheduled slot:** 2026-10-18, 19:30–20:15. **Facilitator / note taker:** Long.

| Member | What have I done since last week? | What will I do until next week? | Issues / problems / obstacles |
|---|---|---|---|
| Long | Organization, risk, schedule, and build-plan drafts are available for review under S2-03/S2-04. The team roles and task baseline provide the basis for coordinating the remaining work. | Complete Spec Kit setup and the constitution (S2-12), coordinate training completion and requirements clarification, resolve Project Plan reviews, and lead the 22 October Review. | Training and setup have missed their original milestones. Their completion and evidence collection must be coordinated with the remaining document and build reviews. |
| Đạt | User/stakeholder descriptions and the PA1 survey baseline are available under S2-06, providing inputs for the mobile flows and workflow diagrams. | Finish the mobile skeleton (S2-18) and two workflows (S2-09), check them against the API and business rules, and review Build 0 and the report before the 22 October Review. | Mobile integration depends on consistent API fields and agreed offline, settlement, and currency behavior. Workflow and integration checks remain to be completed. |
| Nhật | Introduction and product-perspective drafts are available. The initial PA2 report was committed on 16 October as 42b3819. | Complete the backend/database/API skeleton (S2-19), align the mobile and AI contracts, review the Spec Kit output, and support Build 0 verification before the 22 October Review. | Exact money representation, duplicate prevention, and personal/group boundaries must remain consistent across components; the integrated skeleton still needs its Build 0 check. |
| Khiêm | Repository preparation and the Spec Kit/training workstream have been identified for follow-up, with task owners and reviewers recorded in the plan. | Complete the NFR verification methods (S2-10), collect all five training records, coordinate Build 0 checks (S2-26), and prepare Jira/Git and submission evidence for the Review. | Training evidence is not yet complete. Build verification depends on the component owners delivering runnable skeletons, leaving a short window for integrated checks and corrections. |
| Hòa | Vision introduction, positioning, and product-feature drafts are available under S2-05/S2- | Complete feature revisions and the AI-service skeleton (S2-20), align the interface with the backend, check provider-failure handling, and consolidate all | Provider quotas, timeouts, and invalid responses remain integration risks. Individual AI histories |

| Member | What have I done since last week? | What will I do until next week? | Issues / problems / obstacles |
|---|---|---|---|
| | 08 for refinement and peer review. | members' AI declarations before the Review. | must be collected and checked before the shared disclosure is finalized. |

**Follow-up targets:** Complete training by 20 October, reconcile setup and document work before Review, and assess Build 0 and all assigned sprint outputs at the 22 October Review.

**Summary:** The 18 October Scrum focused on completing training and setup, finalizing document reviews, and preparing the integrated skeleton and submission evidence. These workstreams were subsequently completed and accepted at the 22 October Sprint Review.

### 2.4 Sprint Review & Retrospective — 22 October 2026

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

**Date / time:** 2026-10-22, 19:30–20:30.
**Leader / facilitator / note taker:** Phạm Định Tiểu Long.
**Members present:** Phạm Định Tiểu Long, Nguyễn Phú Đạt, Nguyễn Lê Đức Nhật, Vương Đắc Gia Khiêm, and Ngô Thái Hòa.
**Members absent:** None.

**Review goal:** Evaluate the completed PA2 outputs against the sprint goal, confirm submission readiness, and agree on practical improvements for Sprint 3.

#### Review agenda

| Time | Activity | Presenter / lead |
|---|---|---|
| 19:30–19:40 | Present the Project Plan, Vision, workflows, and task consistency checks | Long, Hòa, Đạt, and Nhật |
| 19:40–19:55 | Demonstrate Build 0 and review the recorded startup/connectivity checks | Khiêm with Đạt, Nhật, and Hòa |
| 19:55–20:05 | Review Spec Kit artifacts and the five individual training records | Long, Khiêm, and all members |
| 20:05–20:15 | Review AI disclosures, available meeting records/evidence, the draft PDF, and the final packaging plan | Hòa, Khiêm, and Long |
| 20:15–20:30 | Discuss retrospective findings and agree on Sprint 3 improvements | All members |

#### Deliverable review

| Deliverable | Presenter / owner | Review outcome |
|---|---|---|
| Project Plan | Long; Nhật for the overview | Accepted. Scope, organization, risks, task ownership, separate reviewers, capacity, and staged build milestones were checked for consistency. |
| Vision Document and workflows | Hòa, Đạt, and Nhật | Accepted. The PA1 survey was retained; product features and workflows consistently describe offline support, group accounting, settlement confirmation, and VND/USD conversion. |
| Non-functional requirements | Khiêm | Accepted. Quality targets have measurable criteria and proposed verification methods; target values are distinguished from observed build results. |
| Spec Kit and individual training | Long for setup; Khiêm for evidence collection; each member for their own training | Accepted at the 22 October Review. Individual training and the completed constitution are documented in Part C. |
| Build 0 | Khiêm with the component owners | Accepted for the skeleton milestone. The acceptance scope is startup, mobile/API/database connectivity, AI-service response, and controlled provider-failure handling; later feature and NFR tests belong to subsequent builds. |
| AI Usage Report and Weekly Report | Hòa and Long | Completed for the 22 October Review. The report includes four meeting records and 65 AI usage entries dated on or before 22 October. |
| Evidence and submission readiness | Khiêm, with Đạt and Long reviewing | Assigned evidence and package-preparation work completed by the 22 October Review. |

**Review decision:** At the 22 October Review, the team confirmed completion of all assigned Sprint 2 work: documents, Spec Kit and training, Build 0, and evidence/package preparation. No assigned Sprint 2 work remained outstanding. Feature implementation and quantitative testing continue in Sprints 3–5.

**Retrospective**

| Required question | Discussion and conclusion |
|---|---|
| What went well? | All five members participated in each meeting, and the role-based allocation gave each workstream a clear owner and a different reviewer. Reusing the PA1 baseline helped the team refine the plan and vision, while the shared Build 0 checkpoint brought the component owners together before the final Review. |
| What went wrong? | Spec Kit and individual training missed their initial early-sprint targets. Although the team recovered the work before the Sprint Review, this reduced the time |

| Required question | Discussion and conclusion |
|---|---|
| | available for review and concentrated evidence/package preparation near the end of the sprint. |
| What problems occurred and what caused them? | The team underestimated the combined effort for learning, environment setup, and retaining individual evidence. Component work also needed agreement on API fields and financial rules. Recovery used named owners, revised checkpoints, and a shared startup review to resolve these dependencies. |
| What can be done differently next sprint? | Define each task's acceptance criteria and dependencies during Planning, validate the environment early, and retain evidence when work is completed. Review API examples before dependent feature work and schedule an integration rehearsal before the release build. |
| What lessons were learned? | Training, setup, implementation, review, and evidence collection each require explicit time. Recovery dates should remain traceable alongside the original plan. A shared build checkpoint and concrete accounting examples help the team find cross-component inconsistencies before the final Review. |

**Agreed Sprint 3 improvement actions**

| Improvement | Owner | Reviewer | Target | Success criterion |
|---|---|---|---|---|
| Prepare Sprint 3 tasks with an assignee, different reviewer, estimate, dependencies, and acceptance criteria | Long | Khiêm | 2026-10-26 | Every task selected at Planning contains the required fields before work starts |
| Use a per-task evidence and review checklist from the start of the sprint | Khiêm | Long | 2026-10-26 | A completed sample shows where the output, review decision, and supporting evidence are retained |
| Publish reviewed mobile/backend/AI contract examples, including money and error behavior | Nhật | Đạt | 2026-10-27 | Component owners use the same request/response and accounting examples |
| Run a shared integration rehearsal before Build 1 | Khiêm | Nhật | 2026-11-05 | The rehearsal records results and assigns any defects before the 8 November build target |

**Completion and handoff — 22 October:** All assigned Sprint 2 work was completed by the Review.

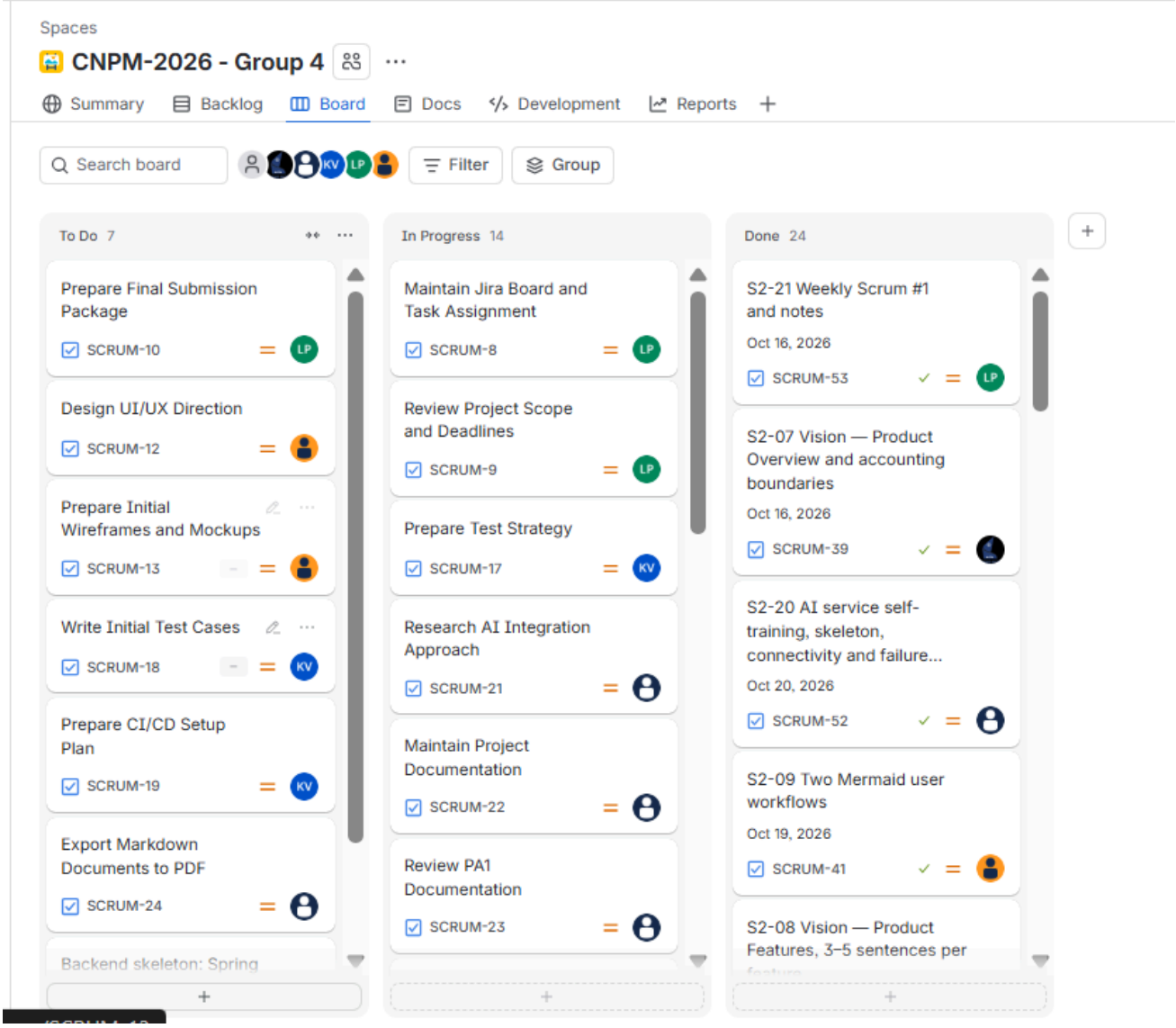
**Summary:** The Sprint Review took place on 22 October. The team reported all assigned Sprint 2 work complete and retained the following-sprint improvement actions above.

**2.5 Jira Evidence**

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

Figure D.1 provides the single Jira board image used for PA2 assignment/progress evidence. The visible project is **CNPM-2026 - Group 4**. Each displayed task card includes a Jira key, title, and assignee avatar, and its column identifies its current status.

**Figure D.1 — Jira board overview for PA2**




*Caption: CNPM-2026 - Group 4 board, showing 7 To Do, 14 In Progress, and 24 Done tasks at the recorded snapshot. The board includes PA2 tasks identified by their S2 prefixes. Evidence image saved locally on 19 October 2026 at 11:33:00 (UTC+07:00).*

The counters cover the displayed project board, including other project tasks; they are not a count of only the 27 Sprint 2 tasks. The following local IDs and Jira keys can be read directly from the visible cards:

| Local ID in A.4.3 | Jira key | Visible task | Status shown |
|---|---|---|---|
| S2-07 | SCRUM-39 | Vision — Product Overview and accounting boundaries | Done |

| Local ID in A.4.3 | Jira key | Visible task | Status shown |
|---|---|---|---|
| S2-09 | SCRUM-41 | Two Mermaid user workflows | Done |
| S2-20 | SCRUM-52 | AI service self-training, skeleton, connectivity and failure checks | Done |
| S2-21 | SCRUM-53 | Weekly Scrum #1 and notes | Done |

### 2.6 Git History Evidence

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

The five screenshots show repository history across branches using `git log --all --graph`, including commit hashes, authors, dates, and messages. The latest visible commit is `40f5263`, dated 21 October 2026. Some screenshots overlap.

```
PS D:\FinSync\FinSync> git log --all --graph --pretty=format:"%h | %an | %ad | %s" --date=short
  * 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
  * 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
  * 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
  * 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
  * 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
  * 8af0078 | Nguyen Nhat | 2026-10-14 | prepare for PA2
  * 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
  * 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
  * 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
  * 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
  * 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
  * 8af0078 | Nguyen Nhat | 2026-10-14 | prepare for PA2
  * 0b77296 | Long Pham | 2026-10-05 | add document
  | * 04c6c2f | Khiem Vuong | 2026-10-20 | docs: Test Strategy
  | * b98938f | Seadarkness | 2026-10-19 | chore: initialize Spec Kit in src
  | * 9194184 | Nguyen Nhat | 2026-10-15 | remove screenshots folder
  | * ddd7d3f | Nguyen Nhat | 2026-10-15 | update .gitignore
  | * 7857c75 | Nguyen Nhat | 2026-10-14 | refactor: reorganize project structure
  |/
  | * 6c5cd20 | Long Pham | 2026-10-19 | add: new speckit
  | * 0343872 | Long Pham | 2026-10-05 | add README
  | * dcf4871 | Long Pham | 2026-10-05 | add new structure
  |/
  * 73ce2cb | Nguyen Nhat | 2026-10-05 | docs: update team member roles and responsibilities in report and add PDF export with Jira board screenshot
  * 5cf3c73 | Nguyen Nhat | 2026-10-05 | docs: update technology stack and architecture details in project report
  * 18639de | Nguyen Nhat | 2026-10-04 | chore: remove outdated report.pdf file
  * bb30f98 | Nguyen Nhat | 2026-10-04 | docs: add comprehensive non-functional requirements specification and project management reports
  * 84c8bde | Nguyen Nhat | 2026-10-04 | docs: initialize comprehensive project documentation, architecture design, and repository structure
  | * be1f54e | Khiem Vuong | 2026-10-04 | Splitwise screenshots
  | * 298cd0c | Khiem Vuong | 2026-10-03 | Splitwise screenshots
  |/
```

*Figure D.2 — PA2 report updates, evidence additions, and Spec Kit commits.*

```
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
* 8af0078 | Nguyen Nhat | 2026-10-14 | prepare for PA2
* 0b77296 | Long Pham | 2026-10-05 | add document
| * 04c6c2f | Khiem Vuong | 2026-10-20 | docs: Test Strategy
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
* 8af0078 | Nguyen Nhat | 2026-10-14 | prepare for PA2
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
```

*Figure D.3 — PA2 documentation and screenshot updates from 14–21 October 2026.*

```
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
* 8af0078 | Nguyen Nhat | 2026-10-14 | prepare for PA2
* 0b77296 | Long Pham | 2026-10-05 | add document
| * 04c6c2f | Khiem Vuong | 2026-10-20 | docs: Test Strategy
| * b98938f | Seadarkness | 2026-10-19 | chore: initialize Spec Kit in src
| * 9194184 | Nguyen Nhat | 2026-10-15 | remove screenshots folder
| * ddd7d3f | Nguyen Nhat | 2026-10-15 | update .gitignore
| * 7857c75 | Nguyen Nhat | 2026-10-14 | refactor: reorganize project structure
|/
| * 6c5cd20 | Long Pham | 2026-10-19 | add: new speckit
| * 0343872 | Long Pham | 2026-10-05 | add README
| * dcf4871 | Long Pham | 2026-10-05 | add new structure
|/
* 73ce2cb | Nguyen Nhat | 2026-10-05 | docs: update team member roles and responsibilities in report and add PDF export with Jira board screenshot
* 5cf3c73 | Nguyen Nhat | 2026-10-05 | docs: update technology stack and architecture details in project report
* 18639de | Nguyen Nhat | 2026-10-04 | chore: remove outdated report.pdf file
:
```

*Figure D.4 — Branch history showing Spec Kit initialization and test strategy commits.*

```
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
* 8af0078 | Nguyen Nhat | 2026-10-14 | prepare for PA2
* 0b77296 | Long Pham | 2026-10-05 | add document
| * 04c6c2f | Khiem Vuong | 2026-10-20 | docs: Test Strategy
| * b98938f | Seadarkness | 2026-10-19 | chore: initialize Spec Kit in src
| * 9194184 | Nguyen Nhat | 2026-10-15 | remove screenshots folder
| * ddd7d3f | Nguyen Nhat | 2026-10-15 | update .gitignore
| * 7857c75 | Nguyen Nhat | 2026-10-14 | refactor: reorganize project structure
|/
| * 6c5cd20 | Long Pham | 2026-10-19 | add: new speckit
| * 0343872 | Long Pham | 2026-10-05 | add README
| * dcf4871 | Long Pham | 2026-10-05 | add new structure
|/
* 73ce2cb | Nguyen Nhat | 2026-10-05 | docs: update team member roles and responsibilities in report and add PDF export with Jira board screenshot
* 5cf3c73 | Nguyen Nhat | 2026-10-05 | docs: update technology stack and architecture details in project report
* 18639de | Nguyen Nhat | 2026-10-04 | chore: remove outdated report.pdf file
* bb30f98 | Nguyen Nhat | 2026-10-04 | docs: add comprehensive non-functional requirements specification and project management reports
* 84c8bde | Nguyen Nhat | 2026-10-04 | docs: initialize comprehensive project documentation, architecture design, and repository structure
| * be1f54e | Khiem Vuong | 2026-10-04 | Splitwise screenshots
| * 298cd0c | Khiem Vuong | 2026-10-03 | Splitwise screenshots
|/
| * ea86684 | ThaisHoaf | 2026-10-04 | docs: add images to illustrate for Money Lover
|/
* e3a14e0 | Nguyen Nhat | 2026-10-03 | feat: add backend Spring Boot service, PostgreSQL database, and Docker configuration
:
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
```

*Figure D.5 — Earlier project documentation, app-survey screenshots, and backend setup.*

```
PS D:\FinSync\FinSync> git log --all --graph --pretty=format:"%h | %an | %ad | %s" --date=short
* 40f5263 | Nguyen Nhat | 2026-10-21 | update report.md
* 02e260a | Nguyen Nhat | 2026-10-20 | Add sprint 2 screenshots for Git log and JIRA board
* 94af8ca | Nguyen Nhat | 2026-10-19 | draft for PA2 report
* 992b4bf | Nguyen Nhat | 2026-10-19 | add AI-Usage.md
* 42b3819 | Nguyen Nhat | 2026-10-16 | prepare for PA2
* 8af0078 | Nguyen Nhat | 2026-10-14 | prepare for PA2
* 0b77296 | Long Pham | 2026-10-05 | add document
| * 04c6c2f | Khiem Vuong | 2026-10-20 | docs: Test Strategy
| * b98938f | Seadarkness | 2026-10-19 | chore: initialize Spec Kit in src
| * 9194184 | Nguyen Nhat | 2026-10-15 | remove screenshots folder
| * ddd7d3f | Nguyen Nhat | 2026-10-15 | update .gitignore
| * 7857c75 | Nguyen Nhat | 2026-10-14 | refactor: reorganize project structure
|/
| * 6c5cd20 | Long Pham | 2026-10-19 | add: new speckit
| * 0343872 | Long Pham | 2026-10-05 | add README
| * dcf4871 | Long Pham | 2026-10-05 | add new structure
|/
* 73ce2cb | Nguyen Nhat | 2026-10-05 | docs: update team member roles and responsibilities in report and add PDF export with Jira b
oard screenshot
* 5cf3c73 | Nguyen Nhat | 2026-10-05 | docs: update technology stack and architecture details in project report
* 18639de | Nguyen Nhat | 2026-10-04 | chore: remove outdated report.pdf file
* bb30f98 | Nguyen Nhat | 2026-10-04 | docs: add comprehensive non-functional requirements specification and project management rep
orts
* 84c8bde | Nguyen Nhat | 2026-10-04 | docs: initialize comprehensive project documentation, architecture design, and repository st
ructure
| * be1f54e | Khiem Vuong | 2026-10-04 | Splitwise screenshots
| * 298cd0c | Khiem Vuong | 2026-10-03 | Splitwise screenshots
|/
| * ea86684 | ThaisHoaf | 2026-10-04 | docs: add images to illustrate for Money Lover
|/
* e3a14e0 | Nguyen Nhat | 2026-10-03 | feat: add backend Spring Boot service, PostgreSQL database, and Docker configuration
```

*Figure D.6 — Repository commit graph covering project setup and PA2 work, 3–21 October 2026.*

### 2.7 Meeting Action Register

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Nguyen Le Duc Nhat`

All actions below were closed by the 22 October Review. Baseline due dates show the original plan.

| Action / source | Action and expected result | Owner | Reviewer | Baseline due | Disposition and closeout milestone |
|---|---|---|---|---|---|
| ACT-01 / Planning | Establish task ownership, dates, capacity, and Jira traceability | Long | Khiêm | 2026-10-10 | Closed by the 22 October Review. |
| ACT-02 / Planning | Complete Spec Kit setup and coordinate five training records | Long | Nhật | 2026-10-15 | Closed by the 22 October Review. |
| ACT-03 / Scrum #1 | Reconcile the baseline and recovery milestones | Long | Hòa | 2026-10-17 | Closed by the 22 October Review. |
| ACT-04 / Scrum #1 | Settle currency, own-group-share, and duplicate-counting rules | Nhật | Khiêm | 2026-10-17 | Closed by the 22 October Review. |
| ACT-05 / Scrum #2 | Assess Build 0 against its skeleton scope | Khiêm | Đạt | 2026-10-22 | Closed by the 22 October Review. |
| ACT-06 / Scrum #2 | Consolidate AI declarations and supporting histories | Hòa | Long | 2026-10-23 | Closed by the 22 October Review. |
| ACT-07 / Review | Reconcile report content and Review decisions | Đạt | Hòa | 2026-10-24 | Closed by the 22 October Review. |
| ACT-08 / Review | Prepare and check the Markdown/PDF submission package | Khiêm | Long | 2026-10-25 | Closed by the 22 October Review. |

# Appendix — AI Usage Notes

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

The 65 entries below are grouped by member. Requested AI contributions describe the assistance sought. Member-reported work is identified separately

## 1. Nguyễn Lê Đức Nhật

> *Performed by:* `Nguyen Le Duc Nhat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Le Duc Nhat`

- **Prompt 1:** Codex(gpt-6.1-sol), Open AI, accessed 7:00 on 10/10/2026, prompt: “Read `pa2_project_assignment_specification.md` and create a table listing a checklist of all deliverables for Parts A, B, C, and D, including section names, formats, required evidence, and submission instructions.”; used to request assistance for Parts A–D; requested AI contribution: a deliverable and submission checklist; student contribution: checked the checklist against the

original specification, added the missing items and deadlines, and used the final version to track the team's progress.

- **Prompt 2:** Codex(gpt-6.1-sol), Open AI, accessed 15:30 on 11/10/2026, prompt: “Based on `GEMINI.md`, write a Project Overview for FinSync in English using one or two paragraphs, a simple academic tone, and no more than 150 words.”; used to request assistance for A.1–A.2 (Introduction and Project Overview); requested AI contribution: an English overview based on `GEMINI.md`; student contribution: compared the draft with `GEMINI.md`, rewrote unclear sentences in our own words, and confirmed the final text is within the 150-word limit.

- **Prompt 3:** Codex(gpt-6.1-sol), Open AI, accessed 16:00 on 11/10/2026, prompt: “List at least 10 risks for the FinSync project covering technical issues, staffing, scheduling, AI, and security. Rate Probability and Impact on a scale of 1–5 and calculate the Risk Score as Probability × Impact in a Markdown table.”; used to request assistance for A.3.2 (Risk Management); requested AI contribution: candidate risks and probability × impact scores; student contribution: removed duplicate or unrealistic risks, adjusted the scores to match the team's real situation, and wrote the mitigation actions.

- **Prompt 4:** Codex(gpt-6.1-sol), Open AI, accessed 19:00 on 13/10/2026, prompt: “Create a Performed/Reviewed/Edited assignment table for all sections, ensuring that the performer and reviewer are different people and rotating the assignments evenly among the five members.”; used to request assistance for the attribution lines across Parts A–D and the role/task tables in A.3.1 and A.4.3; requested AI contribution: a proposed Performed/Reviewed/Edited allocation; student contribution: checked that no performer reviews their own section, adjusted the allocation to each member's role and workload, and confirmed the final table with the team.

- **Prompt 5:** Codex(gpt-6.1-sol), Open AI, accessed 17:00 on 13/10/2026, prompt: “Write a Problem Statement and a Product Position Statement for FinSync using standard templates, based on Section 1 of `GEMINI.md`.”; used to request assistance for B.2.1–B.2.2 (Problem and Product Position Statements); requested AI contribution: draft statements based on GEMINI.md; student contribution: verified the statements against Section 1 of `GEMINI.md`, edited the wording to fit the template, and removed claims outside the project scope.

- **Prompt 6:** Codex(gpt-6.1-sol), Open AI, accessed 8:30 on 15/10/2026, prompt: “Compare FinSync with Money Lover and Splitwise in a table using the following criteria: personal wallets, group expense splitting, AI, budgeting, and offline support.”; used to request assistance for B.3.5 (Alternatives and Competition); requested AI contribution: a comparison table for FinSync, Money Lover, and Splitwise; student contribution: checked each competitor feature against the official app pages and store listings, and corrected the cells that were inaccurate or outdated.

- **Prompt 7:** Codex(gpt-6.1-sol), Open AI, accessed 9:10 on 16/10/2026, prompt: “Write an 80–100-word description in English of Feature 6: Smart Debt Split & Settlement, based on `GEMINI.md`.”; used to request assistance for B.5.6 (Smart Debt Split & Settlement); requested AI contribution: an English feature-description draft; student contribution: compared the draft with the feature specification in `GEMINI.md`, rewrote it to match our design, and confirmed the word count is 80–100.

- **Prompt 8:** Codex(gpt-6.1-sol), Open AI, accessed 7:45 on 17/10/2026, prompt: “Draw a Mermaid flowchart showing the following workflow: record a shared expense → split the amount → send notifications → mark as paid.”; used to request assistance for B.5.12 (Key Workflow Diagrams); requested AI contribution: Mermaid syntax for recording, splitting, notifying, and settling shared expenses; student contribution: rendered the diagram to test it, fixed syntax errors, and added decision branches so the flow matches the real behavior of the app.

- **Prompt 9:** Codex(gpt-6.1-sol), Open AI, accessed 7:45 on 20/10/2026, prompt: “Propose Non-Functional Requirements covering Performance, Security, Privacy, Usability, and Reliability in a table with the columns ID, Category, Requirement, Measurement, and Priority.”; used to request assistance for B.6 (Non-Functional Requirements); requested AI contribution: candidate requirements, measurement criteria, and priorities; student contribution: selected the requirements relevant to FinSync, changed the measurements into realistic and testable targets, and set the final priorities with the team.

- **Prompt 10:** Codex(gpt-6.1-sol), Open AI, accessed 13:30 on 20/10/2026, prompt: “Briefly explain the Spec Kit workflow (constitution → specify → plan → tasks → implement) and provide instructions for initializing it in the existing FinSync repository.”; used to request assistance for C.1 (Repository and Spec Kit Setup) and C.4 (Training Evidence); requested AI contribution: a workflow explanation and suggested initialization steps; student contribution: ran the initialization in the real FinSync repository, fixed any setup errors, and saved the commands and screenshots as evidence.

- **Prompt 11:** Codex(gpt-6.1-sol), Open AI, accessed 7:45 on 21/10/2026, prompt: “Draft a constitution prompt for Spec Kit based on the coding conventions in Section 6.6 of `GEMINI.md`.”; used to request assistance for C.3 (Constitution Summary); requested AI contribution: a constitution prompt based on GEMINI.md Section 6.6; student contribution: compared the prompt with Section 6.6, edited it to match the team's real conventions, ran it in Spec Kit, and checked the generated constitution.

- **Prompt 12:** Codex(gpt-6.1-sol), Open AI, accessed 8:35 on 21/10/2026, prompt: “I am the Backend Lead working with Spring Boot. Explain how to use Spec Kit for my role and provide a small exercise for independent practice.”; used to request assistance for C.4 (Spec Kit Training Evidence); requested AI contribution: a backend-specific explanation and a Spring Boot practice exercise; student contribution: completed the exercise independently, wrote personal notes on the Backend Lead workflow, and recorded the result as training evidence.

- **Prompt 13:** Codex(gpt-6.1-sol), Open AI, accessed 14:25 on 21/10/2026, prompt: “Create a checklist to verify that the Spec Kit files have been committed, `.gitignore` is configured correctly, and no API keys or `.env` files are exposed.”; used to request assistance for C.1–C.2 (Repository Setup and Submitted Artifacts); requested AI contribution: a checklist for tracked Spec Kit files, ignore rules, and possible secret exposure; student contribution: ran the checks manually on the repository (`git status`, `git ls-files`, searching for secrets), fixed the issues found, and confirmed no sensitive files were committed.

- **Prompt 14:** Codex(gpt-6.1-sol), Open AI, accessed 19:30 on 21/10/2026, prompt: “Suggest a structure for the AI Usage Report covering policy, logs, a summary by report part, validation, risks, and reflection.”; used to request assistance for D.1 (AI Usage Report) and the AI Usage Notes appendix; requested AI contribution: a proposed structure for policy, logs, affected sections,

verification, risks, and reflection; student contribution: chose the final structure, filled it in with the real prompt logs, and wrote the validation and reflection sections from our own experience.

### 2. Phạm Định Tiểu Long

> *Performed by:* `Pham Dinh Tieu Long` | *Reviewed by:* `Nguyen Le Duc Nhat` | *Edited by:* `Pham Dinh Tieu Long`

- **Prompt 1:** Claude(3-opus), Anthropic, accessed 08:30 on 10/10/2026, prompt: "Based on the FinSync requirements, propose a Domain-Driven Design (DDD) package structure for a Modular Monolith in Spring Boot, specifically defining bounded contexts for User Management, Group Expenses, and AI Financial Advisor."; used to request assistance with backend domain boundaries related to B.4.1 (Product Perspective); requested AI contribution: a DDD package structure for a modular Spring Boot backend; student validation: mapped the proposed boundaries to the personal/group bookkeeping requirements and kept the FastAPI AI service separate as shown in B.4.1.

- **Prompt 2:** Gemini(1.5-pro), Google, accessed 14:15 on 11/10/2026, prompt: "Write a DBML code defining the database schema for FinSync in PostgreSQL. Include tables for `Users`, `Groups`, `Expenses`, and `Settlements`, and add `sync_status` and `last_updated` columns to support offline-first synchronization."; used to explore data relationships supporting B.5.4 (Personal Transaction Tracking), B.5.5 (Group Management & Shared Wallets), and B.5.6 (Smart Debt Split & Settlement); requested AI contribution: a draft DBML schema; student validation: checked entity relationships, normalization, record ownership, and synchronization fields against B.4.1 before adopting the schema.

- **Prompt 3:** Claude(3-opus), Anthropic, accessed 09:45 on 12/10/2026, prompt: "Draft a REST API contract for the Expense module in FinSync. Include endpoints for recording shared expenses, updating offline transactions, and retrieving group balances, specifying HTTP methods and JSON payloads."; used to request API design assistance related to B.5.4 (Personal Transaction Tracking), B.5.6 (Smart Debt Split & Settlement), and B.6.6 (Maintainability and Quality, NFR-M3); requested AI contribution: endpoint and JSON payload proposals; student validation: checked authorization, synchronization identifiers, error responses, and the requirement that shared ledger changes occur online before matching the contract to implemented DTOs.

- **Prompt 4:** Gemini(1.5-pro), Google, accessed 20:00 on 13/10/2026, prompt: "Write a Mermaid sequence diagram showing the backend logic when a user submits an offline expense batch to the Spring Boot server, including conflict resolution and database updates."; used to explore synchronization behavior related to B.4.1 (Product Perspective), B.5.4 (Personal Transaction Tracking), and B.6.3 (Reliability, Offline Operation, and Financial Correctness, NFR-R6); requested AI contribution: a draft Mermaid synchronization sequence; student validation: restricted queued offline writes to personal transactions, retained both conflicting versions for explicit user resolution, and required online access for shared ledger changes.

- **Prompt 5:** Claude(3-opus), Anthropic, accessed 15:20 on 14/10/2026, prompt: "Provide an example of configuring Spring Data Redis to cache user group balances, including the CacheManager setup and `@Cacheable` annotations for repository methods."; used to explore a caching option in relation to B.6.1 (Performance); requested AI contribution: an example Spring

Data Redis configuration; student validation: assessed consistency, invalidation, operational cost, and measured performance before adopting caching. Redis is an exploratory option outside the baseline in B.4.1 and B.4.2.

- **Prompt 6:** Gemini(1.5-pro), Google, accessed 10:10 on 15/10/2026, prompt: "Write a Java algorithm using a greedy approach (min-cash flow) to optimize and settle debts among a group of users who owe each other different amounts."; used to request algorithm assistance related to B.5.6 (Smart Debt Split & Settlement) and B.6.3 (Reliability, Offline Operation, and Financial Correctness); requested AI contribution: an example debt simplification algorithm; student validation: used exact monetary arithmetic, preserved each member's net balance, applied the agreed rounding rules, and tested representative cases before integrating the algorithm.

- **Prompt 7:** Claude(3-opus), Anthropic, accessed 16:45 on 16/10/2026, prompt: "Generate a UML Class Diagram in Mermaid syntax for the JPA Entities: `User`, `ExpenseGroup`, `Transaction`, and `DebtBalance`, showing their cardinalities (OneToMany, ManyToMany)."; used to explore entity relationships supporting B.5.4 (Personal Transaction Tracking), B.5.5 (Group Management & Shared Wallets), and B.5.6 (Smart Debt Split & Settlement); requested AI contribution: draft Mermaid class-diagram syntax; student validation: checked cardinalities, ownership, and ledger relationships against the project data model and verified rendering.

- **Prompt 8:** Gemini(1.5-pro), Google, accessed 09:00 on 17/10/2026, prompt: "Explain how to implement stateless authentication using Spring Security 6 and JWT. Provide the `SecurityFilterChain` configuration required to secure FinSync's API endpoints."; used to request implementation guidance related to B.5.1 (Authentication & Security) and B.6.2 (Security); requested AI contribution: an example JWT and Spring Security configuration; student validation: reviewed public and protected routes, validated tokens, and tested access to personal and group records against the stated authorization requirements.

- **Prompt 9:** Claude(3-opus), Anthropic, accessed 14:30 on 18/10/2026, prompt: "Design a communication interface (using RestTemplate or WebClient) for the Spring Boot backend to send user spending data to the external Python AI Financial Advisor service."; used to request integration guidance related to B.4.1 (Product Perspective) and B.5.11 (AI Financial Advisor); requested AI contribution: example Java client code for the FastAPI service; student validation: evaluated the client choice, verified request/response serialization, and tested timeouts and controlled provider-failure behavior before adopting the implementation.

- **Prompt 10:** Gemini(1.5-pro), Google, accessed 19:15 on 19/10/2026, prompt: "Write a global `@RestControllerAdvice` class in Spring Boot to handle custom exceptions like `GroupNotFoundException` and `InsufficientBalanceException`, returning a standard `ErrorResponse` DTO."; used to request error-handling guidance related to B.6.6 (Maintainability and Quality, NFR-M3); requested AI contribution: exception-handler boilerplate and an error-response DTO; student validation: aligned validation fields, timestamps, and status codes with the API contract, avoided exposing sensitive details, and tested documented error responses.

- **Prompt 11:** Claude(3-opus), Anthropic, accessed 08:45 on 20/10/2026, prompt: "Write an optimized PostgreSQL index creation script for queries that frequently filter expenses by `group_id` and sort them by `created_at` in descending order."; used to explore query optimization related to B.6.1 (Performance) and the PostgreSQL baseline in B.6.5 (Platform and Compatibility); requested

AI contribution: candidate SQL indexes; student validation: compared EXPLAIN ANALYZE output before and after each index using representative data and recorded the results before claiming a performance improvement.

- **Prompt 12:** Gemini(1.5-pro), Google, accessed 15:00 on 21/10/2026, prompt: "Create a `docker-compose.yml` file for a local development environment that includes a Spring Boot app container, a PostgreSQL 15 database, and a Redis container. Include health checks and network bridges."; used to request environment configuration guidance related to A.4.4 (Build Plan) and B.6.5 (Platform and Compatibility, NFR-C3); requested AI contribution: an example Docker Compose configuration; student validation: adapted the requested PostgreSQL 15 example to the report's PostgreSQL 16 baseline, omitted Redis unless separately approved, and checked health checks, networking, external configuration, and startup.

- **Prompt 13:** Claude(3-opus), Anthropic, accessed 10:30 on 22/10/2026, prompt: "Write a parameterized JUnit 5 test using `@CsvSource` to verify the automated expense splitting logic for different input amounts, participant counts, and split ratios (equal, percentage, exact)."; used to plan financial test cases related to B.5.6 (Smart Debt Split & Settlement), B.6.6 (Maintainability and Quality, NFR-M1 and NFR-M6), and B.6.7 (Standards and Verification); requested AI contribution: parameterized JUnit 5 test examples; student validation: derived expected values independently, included rounding and split-total invariants, and retained actual local or CI test results.

- **Prompt 14:** Gemini(1.5-pro), Google, accessed 14:20 on 22/10/2026, prompt: "Draft a short technical justification explaining why we chose a Modular Monolith over Microservices for FinSync, citing team size, domain complexity, and deployment overhead."; used to request architecture rationale assistance for B.4.1 (Product Perspective); requested AI contribution: trade-off analysis for a modular backend; student validation: evaluated the reasoning against team size, deployment constraints, and the separate FastAPI service, then wrote the final rationale in the student's own words.

- **Prompt 15:** Claude(3-opus), Anthropic, accessed 20:30 on 22/10/2026, prompt: "Write a personal reflection on how using AI tools assisted in designing the database schema, resolving Spring Security dependency issues, and writing complex debt-splitting algorithms."; used to prepare the member's reflection for D.1 (AI Usage Report) and this Appendix; requested AI contribution: a draft reflection about backend AI assistance; student validation: wrote a personal account based on actual usage, identified suggestions adopted or rejected, and described only debugging, algorithm changes, and checks supported by the member's own work.

### 3. Nguyễn Phú Đạt

> *Performed by:* `Nguyen Phu Dat` | *Reviewed by:* `Pham Dinh Tieu Long` | *Edited by:* `Nguyen Phu Dat`

- **Prompt 1:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 08:42 on 10/10/2026, prompt: "Based on PA2 Part C, briefly list the tasks I need to complete for the Spec Kit Initialization section."; used to clarify the required deliverables for Part C; requested AI contribution: a concise checklist of setup, generated artifacts, training evidence, and repository verification; student validation: compared the checklist with the PA2 specification and the team template before carrying out the steps.

- **Prompt 2:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 14:17 on 10/10/2026, prompt: "The FinSync repository already contains `/src/admin-web`, `/src/backend`, and `/src/mobile`. Where should I initialize Spec Kit to match the assignment requirements?"; used to determine the correct setup location; requested AI contribution: explain how to integrate Spec Kit into the existing repository without changing the project structure; student validation: verified that the generated Spec Kit files appear under `/src`.

- **Prompt 3:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 19:05 on 11/10/2026, prompt: "Guide me through installing Spec Kit on Windows using VS Code and PowerShell, with short step-by-step instructions."; used to prepare the local development environment; requested AI contribution: installation steps for `uv`, Python, and Spec Kit CLI; student validation: executed each command locally and verified the resulting versions.

- **Prompt 4:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 09:31 on 12/10/2026, prompt: "The `specify version` command is failing on my machine. How should I check Python and reinstall Spec Kit?"; used to troubleshoot the Spec Kit CLI; requested AI contribution: identify a likely runtime-version issue and suggest a clean reinstall procedure; student validation: tested the suggested Python version and reran `specify version`.

- **Prompt 5:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 16:26 on 13/10/2026, prompt: "I have already installed the Spec Kit CLI. For a non-empty `/src` directory, which initialization command should I use if the team uses GitHub Copilot?"; used to initialize Spec Kit in the existing source tree; requested AI contribution: suggest an initialization command for a non-empty directory and Copilot integration; student validation: ran the command inside `/src` and inspected the generated directories before committing anything.

- **Prompt 6:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 10:14 on 15/10/2026, prompt: "After initialization, which files and folders should I check to confirm that Spec Kit was set up successfully?"; used to verify the initialization; requested AI contribution: a short verification checklist covering `.specify`, `.github`, `constitution.md`, templates, and scripts; student validation: inspected the actual repository structure and recorded screenshots.

- **Prompt 7:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 15:48 on 16/10/2026, prompt: "Give me the PowerShell commands to find `constitution.md` and list all Markdown files generated by Spec Kit."; used to collect artifact evidence; requested AI contribution: PowerShell commands for locating the constitution and generated Markdown files; student validation: compared the command output with the files visible in the repository.

- **Prompt 8:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 20:11 on 17/10/2026, prompt: "After verifying the Spec Kit setup, which files should I commit and what should I check before pushing to GitHub?"; used to prepare the repository update; requested AI contribution: a safe Git workflow for reviewing and committing the generated Spec Kit files; student validation: inspected `git status`, reviewed changed files, then committed and pushed only the intended artifacts.

- **Prompt 9:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 09:22 on 19/10/2026, prompt: "How can I verify on GitHub that the Spec Kit Initialization was pushed successfully and that I have enough evidence for the report?"; used to verify the remote repository and evidence; requested AI contribution: a checklist for checking the `.specify` folder, `constitution.md`, generated

templates, and the corresponding Git commit; student validation: opened the repository and compared the remote state with the local setup.

- **Prompt 10:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 18:07 on 20/10/2026, prompt: “Summarize Tutorial #6 about the `/tasks` command in a concise student-style paragraph that is suitable for training evidence.”; used for C.4 (Spec Kit Training Evidence); requested AI contribution: a concise learning summary explaining `tasks.md`, task IDs, dependencies, user-story grouping, and parallelizable tasks; student validation: compared the summary with the tutorial content and the screenshot taken during training.

- **Prompt 11:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 13:36 on 21/10/2026, prompt: “Which screenshots should I capture as evidence for the Spec Kit setup and Tutorial #6?”; used to prepare report evidence; requested AI contribution: suggest a minimal screenshot set showing initialization, constitution path, generated Markdown artifacts, Git push, and training completion; student validation: captured the screenshots directly from the actual environment.

- **Prompt 12:** ChatGPT (gpt-6.1-sol), OpenAI, accessed at 21:04 on 21/10/2026, prompt: “Based on the evidence I already have, help me fill in the relevant fields in the Part C template while keeping the team’s original file structure unchanged.”; used to organize the final Part C documentation; requested AI contribution: map verified setup information and evidence paths into the existing template; student validation: checked all names, dates, paths, statuses, and evidence links against the actual repository before submission.

### 4. Vương Đắc Gia Khiêm

> *Performed by:* `Vuong Dac Gia Khiem` | *Reviewed by:* `Ngo Thai Hoa` | *Edited by:* `Vuong Dac Gia Khiem`

- **Prompt 1:** ChatGPT (GPT-6), OpenAI, accessed 8:38 on 14/10/2026, prompt: “I am preparing a Test Strategy for the FinSync project. Is the following structure complete and suitable for a beginner: (1) Test Objectives, (2) Scope, (3) Test Environment, (4) Testing Methods & Types (manual/automation, functional/performance/security), (5) Tools, and (6) Risk Analysis & Contingency Plan?”; used to request assistance with planning the Test Strategy document; requested AI contribution: review the proposed structure and recommend missing sections; student validation: verified that the final structure matches the project requirements, the team's expectations, and the actual repository setup.

- **Prompt 2:** ChatGPT (GPT-6), OpenAI, accessed 9:40 on 14/10/2026, prompt: “Rewrite a Markdown file based on your recommendations for the FinSync Test Strategy.”; used to request assistance with preparing the Test Strategy document; requested AI contribution: draft TEST_STRATEGY.md with sections covering objectives, scope, environment, testing methods and types, tools and frameworks, test prioritization, entry and exit criteria, CI/CD integration and reporting, and risk analysis and contingency planning; student validation: checked the document against the actual project architecture, implemented features, installed testing tools, build commands, and CI/CD configuration, and revised any assumptions that had not been verified.

- **Prompt 3:** ChatGPT (GPT-6), OpenAI, accessed 13:20 15/10/2026, prompt: “Explain each section of the Test Strategy so that I can understand it and avoid becoming dependent on AI.”; used to

request assistance with understanding the Test Strategy and learning the underlying testing concepts; requested AI contribution: explain the purpose and practical application of each section, including test objectives, scope, environment, testing methods and levels, tools, prioritization, entry and exit criteria, CI/CD integration, and risk analysis; student validation: explained the concepts in their own words, related them to actual FinSync requirements and code, and independently justified the selected testing methods, priorities, and completion criteria.

- **Prompt 4:** ChatGPT (GPT-6), OpenAI, accessed 17:54 15/10/2026, prompt: “Explain the difference between a Test Strategy, Test Plan, Test Case, and Test Scenario using simple examples from a software project.”; used to clarify fundamental testing concepts before preparing the FinSync Test Strategy; requested AI contribution: definitions, comparisons, and practical examples; student validation: explained each concept in their own words and verified that the distinctions match the course materials.

- **Prompt 5:** ChatGPT (GPT-6), OpenAI, accessed 10:22 16/10/2026, prompt: “Explain how automated tests can be integrated into a CI/CD pipeline and how test results, failures, and reports should be handled. Keep the recommendations independent of any specific CI platform.”; used to develop the CI/CD Integration and Reporting section; requested AI contribution: a general pipeline sequence and reporting recommendations; student validation: inspected the repository's workflow configuration and verified which stages, commands, and reports were supported by the actual setup.

- **Prompt 6:** ChatGPT (GPT-6), OpenAI, accessed 19:36 18/10/2026, prompt: “Explain entry criteria, exit criteria, and quality gates in a software testing strategy, and provide examples suitable for a CI/CD pipeline.”; used to define when testing should begin and when a build may be considered ready to proceed; requested AI contribution: sample criteria and explanations of quality gates; student validation: adjusted the criteria to the team's agreed quality standards and the checks that could actually run in its pipeline.

- **Prompt 7:** ChatGPT (GPT-6), OpenAI, accessed 21:39 18/10/2026, prompt: “Explain the difference between manual testing and automated testing, including their advantages, limitations, and appropriate use cases in a CI/CD workflow.”; used to develop the Testing Methods section; requested AI contribution: a comparison and examples of when to use each method; student validation: chose an approach that matches the team's time, skills, and actual CI/CD setup.

- **Prompt 8:** ChatGPT (GPT-6), OpenAI, accessed 15:30 19/10/2026, prompt: “Explain the purpose of the Test Objectives section in a Test Strategy and suggest measurable objectives for a student software project without assuming unverified project requirements.”; used to develop the Test Objectives section; requested AI contribution: example objectives and guidance on making them measurable; student validation: selected only objectives relevant to FinSync and avoided claiming that unverified targets have been achieved.

- **Prompt 9:** ChatGPT (GPT-6), OpenAI, accessed 16:55 19/10/2026, prompt: “Explain the difference between the SpecKit Specify, Clarify, Plan, Tasks, and Implement stages. For each stage, identify its purpose, expected inputs, outputs, and when it should be used.”; used to understand the overall SpecKit workflow; requested AI contribution: a comparison of the stages and their dependencies; student validation: compared the explanation with the available workflow instructions and traced the stages using the project's actual files.

- **Prompt 10:** ChatGPT (GPT-6), OpenAI, accessed 18:11 19/10/2026, prompt: “Review constitution.md and explain its purpose in the SpecKit workflow. Identify the principles, constraints, and development standards that should guide later specification, planning, and implementation tasks.”; used to understand the role of the project constitution in SpecKit; requested AI contribution: an explanation of the constitution and its relationship to later workflow stages; student validation: compared the explanation with the actual constitution file and identified which principles apply to the FinSync project.

- **Prompt 11:** ChatGPT (GPT-6), OpenAI, accessed 12:30 20/10/2026, prompt: “Read the relevant SKILL.md file for the SpecKit Specify workflow and explain its required inputs, steps, outputs, and constraints in beginner-friendly language.”; used to understand how the Specify workflow operates; requested AI contribution: a step-by-step explanation of the skill instructions; student validation: followed the actual instructions and verified that the explanation matches the files and commands in the repository.

- **Prompt 12:** ChatGPT (GPT-6), OpenAI, accessed 13:10 20/10/2026, prompt: “Read the SpecKit Clarify instructions and explain how to identify ambiguities, missing requirements, and important decisions before planning implementation.”; used to understand the Clarify stage; requested AI contribution: an ambiguity checklist and examples of useful clarification questions; student validation: identified genuine unanswered questions in the FinSync specification and distinguished them from assumptions that already have evidence.

- **Prompt 13:** ChatGPT (GPT-6), OpenAI, accessed 14:30 20/10/2026, prompt: “Explain the SpecKit Plan workflow and how its technical plan should be derived from an approved feature specification and the project's existing constitution and technology constraints.”; used to understand technical planning in SpecKit; requested AI contribution: an explanation of planning inputs, technical decisions, and expected outputs; student validation: compared proposed technologies and architecture with the actual repository, existing dependencies, and team decisions.

## 5. Ngô Thái Hòa

> *Performed by:* `Ngo Thai Hoa` | *Reviewed by:* `Vuong Dac Gia Khiem` | *Edited by:* `Ngo Thai Hoa`

- **Prompt 1:** Gemini (gemini-2.5-pro), Google, accessed 06:15 on 10/10/2026, prompt: “I want to install GitHub Spec Kit for my personal project on macOS. The required components are uv, VS Code, Gemini CLI, and specify-cli. I installed specify-cli, but when checking, VS Code status shows 'not found'. What is the cause, and in which directory on macOS should I install specify-cli?”, used to request assistance for environment troubleshooting and PATH configuration; requested AI contribution: diagnostic analysis of macOS PATH resolution and CLI installation best practices; member-reported independent work: manually installed VS Code CLI binary into PATH via Command Palette and verified system execution using `code --version`.

- **Prompt 2:** Gemini (gemini-2.5-pro), Google, accessed 06:25 on 21/10/2026, prompt: “Does installing via uv have global availability similar to npm?”, used to request assistance for package management concepts; requested AI contribution: comparative analysis between `uv tool install` and `npm install -g` regarding execution scope, isolation, and binary pathing; member-reported independent work: verified global executable linkage in `~/.local/bin` while confirming isolated tool virtual environments under `~/.local/share/uv/tools/`.

- **Prompt 3:** Gemini (gemini-2.5-pro), Google, accessed 06:28 on 10/10/2026, prompt: “I ran `uv tool list` and `npm list -g --depth=0` which showed `gemini-cli v0.2.5` and `specify-cli v1.2.1.dev0` under uv, but `zsh: command not found: npm`. What should I do next?”, used to request assistance for CLI environment resolution; requested AI contribution: environment diagnostic interpretation and configuration steps for Zsh shell PATH integration; member-reported independent work: exported `~/.local/bin` to `$PATH` in `~/.zshrc` and verified execution of isolated Python binaries without Node.js dependencies.

- **Prompt 4:** Gemini (gemini-2.5-pro), Google, accessed 06:37 on 10/10/2026, prompt: “Which script type should be used for GitHub Spec Kit when integrating with Gemini CLI?”, used to request assistance for software engineering workflow design; requested AI contribution: architectural recommendation on script runtime selection (Shell/Bash vs Python) for CLI pipeline orchestration; member-reported independent work: selected Shell/Bash as the primary script engine for zero-dependency POSIX pipeline execution across macOS developer workstations.

- **Prompt 5:** Gemini (gemini-2.5-pro), Google, accessed 06:50 on 10/10/2026, prompt: “How should a 5-member team collaborate using GitHub Spec Kit? Is it mandatory to use the same AI tool?”, used to request assistance for Agile team management and AI-assisted workflow strategy; requested AI contribution: team collaboration framework covering Git branching, task distribution, AI model consistency standards, and PR review protocols; member-reported independent work: established Spec Lead roles, feature branch conventions (`feature/*`), and shared repository specification files while enabling flexible individual AI tool usage.

- **Prompt 6:** Claude (Sonnet 5.5), Anthropic, accessed 09:30 on 13/10/2026, prompt: “Is the software architecture above easy to extend? I want to add some AI and machine learning features later to improve the quality of the product and include it in my CV when applying for jobs. I am building it using spec-driven development (SDD) with GitHub Spec Kit.”, used to assess the extensibility of the proposed architecture, identify machine learning features worth adding, and align the development process with Spec Kit;

- **Prompt 7:** Claude (Sonnet 5.5), Anthropic, accessed 09:44 on 13/10/2026, prompt: “Is this process (constitution → specify → clarify → plan → tasks → analyze → implement) repeated for many features, or are all features described at once and the process carried out only a single time?”, used to clarify the methodology of spec-driven development, in particular the scope and iteration of each Spec Kit cycle;

- **Prompt 8:** Claude (Sonnet 5.5), Anthropic, accessed 09:57 on 13/10/2026, prompt: “Please write a sample of the process for 000.”, used to produce a worked example of the full Spec Kit workflow (constitution, specify, clarify, plan, tasks, analyze, implement) for the foundation specification of the project;

- **Prompt 9:** ChatGPT (GPT-6), OpenAI, accessed 07:14 on 21/10/2026, prompt: “Investigate a Gemini CLI session that has been running the GitHub Spec Kit `/speckit.plan` command for approximately 50 minutes without producing new output. Recommend safe diagnostic steps, explain when to interrupt the process, and describe how to check whether partial plan artifacts have already been generated without losing existing project work.”, used to troubleshoot stalled AI-agent execution and preserve partially generated project artifacts;

- **Prompt 10:** ChatGPT (GPT-6), OpenAI, accessed 07:33 on 21/10/2026, prompt: “Analyze the following Gemini CLI error: HTTP 429, `Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests`, with a limit of 20 requests for the `gemini-3-flash` model and a server-recommended retry delay of approximately 8 hours and 34 minutes. Explain the meaning of the error, how free-tier quotas and retry delays affect agent execution, and what troubleshooting options are available.”, used to interpret API quota errors and understand their effects on AI coding workflows;
- **Prompt 11:** ChatGPT (GPT-6), OpenAI, accessed 07:41 on 21/10/2026, prompt: “Evaluate whether GitHub Spec Kit supports Google Antigravity as an alternative to Gemini CLI in Visual Studio Code when Google account authentication is unavailable or unsuccessful. Explain the integration requirements, authentication differences, installation and configuration steps, supported workflow commands, and the considerations involved in continuing an existing Spec Kit project without recreating its specifications or implementation artifacts.”, used to assess compatibility between AI coding environments and determine a practical migration strategy for an existing software project.